



eVTOL UAM Flying Car | A feasibility, opportunity, technology, scaling and risks perspective

Organization Challenger Deep

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AAM's opportunity is real but conditional. Its upside includes new passenger and cargo services, congestion reduction, localized economic development around eVTOL sites, and lower-emission transportation options, as framed in Orlando's AAM planning work. At a broader scale, AAM has been projected as a potential \$500 billion global market by 2040, with logistics applications moving faster than passenger adoption and companies already operating drone delivery services.

AAM should advance through disciplined pilots, not broad deployment. Near-term work should align with statewide transportation plans and focus on the next three to five years, as WSDOT's roadmap does, while treating AAM as a supplement to existing aviation, transit, freight, and energy systems rather than a stand-alone mode. First, prioritize existing aviation assets for early operations.

Executive Summary

Key findings

Urban air mobility is a promising but conditional opportunity: the concept targets passenger, cargo, and urban aviation operations through advanced aircraft integrated with multimodal transportation systems, responding to congestion, pollution, and demand for faster transit in and around cities.¹

The largest gating issues are legal, regulatory, and certification readiness. Current UAM deployment faces unresolved authority questions across federal, state, and local government; challenges span air traffic management, flight standards, environmental policy, public use, land use, and local restrictions. The certification path also remains unclear, with gaps in means of compliance despite ongoing FAA, ASTM, RTCA, SAE, and related efforts.²

Autonomy is both an opportunity and a barrier. The current legal framework does not fully address operations over people, beyond visual line of sight, commercial carriage of cargo or passengers, or airworthiness certification for remote and automated systems; assured autonomy remains a technical and legal challenge. Public acceptance is not assured either: survey findings show stronger preference for piloted operations, some apprehension about flying alone in automated or remotely piloted aircraft, and possible need for mixed fleets or discounts for remote/automated services.²

The most credible early passenger use cases appear to be longer inter-city or inter-regional trips rather than very short urban hops. Survey and focus-group findings indicate preference for routes such as Washington, DC to Baltimore and Los Angeles to San Diego, while very short trips face resistance from cost, transfers, and competition with existing modes on time and price.²

Operations will require robust weather, infrastructure, and grid planning. Adverse weather can reduce or halt operations, extend trip distance, reduce battery life, affect passenger comfort, stress vertiport infrastructure, and disrupt traffic-flow management. Market and planning literature correspondingly treats batteries, charging standards, vertiport deployment, UTM/ATM integration, communications, pilot/workforce demand, and social license as core AAM scaling requirements.^{2,3}

Local economics can be material but site-specific. Orlando's conceptual downtown vertiport analysis used a \$30 million retrofit assumption for Geico Garage, including charging infrastructure, passenger and crew facilities, maintenance facilities, vehicle gates, and structural contingency; it also identified \$600,000 to \$1 million in potential lost annual parking revenue, partly offset by a possible \$2.1 million annual ground lease.⁴

Risks and opportunities

AAM's opportunity is real but conditional. Its upside includes new passenger and cargo services, congestion reduction, localized economic development around eVTOL sites, and lower-emission transportation options, as framed in Orlando's AAM planning work. At a broader scale, AAM has been projected as a potential \$500 billion global market by 2040, with logistics applications moving faster than passenger adoption and companies already operating drone delivery services.⁴⁵

The largest risks sit in the gap between ambition and deployable systems. Low-altitude UAS and ODM aviation bring higher-frequency operations closer to the public, exposing unresolved questions about airspace ownership, regulation, and public acceptance. Current legal summaries show uncertainty below navigable airspace boundaries: above those boundaries, airspace is treated as commons and FAA-regulated, while below them ownership and regulatory authority can remain unclear or shared between FAA and state actors. Later synthesis narrows but does not eliminate the ambiguity, describing airspace below 50 ft as likely private, 50-400 ft as unknown in some contexts, and takeoff-and-landing areas as potentially handled through avigation easements.⁶

Technical risk is also material. NDARC and SUAVE produce broadly similar gross takeoff weights, yet differ sharply in component estimates, with NDARC's cruise lift-to-drag estimate nearly 20% higher and total cruise drag differing by 22%. Across major components, the two methodologies show about a 7% GTOW difference and substantial weight-estimation uncertainty, while battery and motor together account for roughly 40% of aircraft gross takeoff weight. These uncertainties matter commercially because limited battery specific energy reduces aircraft range, making takeoff, climb, descent, motor torque, wiring, and battery sizing central to feasibility rather than secondary engineering details.⁷

Infrastructure is both the binding constraint and the investable opportunity. Orlando's Geico Garage case assumed a \$30 million vertiport retrofit, including charging, passenger and crew facilities, MRO facilities, gates, structural reinforcement contingency, and likely roof-deck reinforcement because the garage was not originally designed for rooftop structures. The same case estimated \$600,000-\$1 million in annual lost parking revenue, but identified a potential \$2.1 million annual ground lease to an AAM operator that could offset that loss. More broadly, policy analysis warns that a mostly private vertiport buildout could limit access, competition, multimodal integration, and public benefit, while open-access public vertiports, interoperable data frameworks, cybersecurity, PPPs, tax incentives, and a national infrastructure fund are positioned as mechanisms to scale the sector.⁴⁵

The practical conclusion is that AAM should be treated neither as speculative hype nor as ready infrastructure. The opportunity is strongest where public agencies can pair high-value use cases with deliberate siting, shared-access infrastructure, legal risk management, and conservative technical assumptions. The central risk is premature scale: deploying before airspace rights, vertiport access, operating economics, aircraft performance, and community impacts are sufficiently resolved.

Recommendations and next steps

AAM should advance through disciplined pilots, not broad deployment. Near-term work should align with statewide transportation plans and focus on the next three to five years, as WSDOT's roadmap does, while treating AAM as a supplement to existing aviation, transit, freight, and energy systems rather than a stand-alone mode.⁸

First, prioritize existing aviation assets for early operations. Airports already have runways, taxiways, procedures, surveillance systems, and in towered settings, controllers who can integrate AAM traffic with conventional operations. But reuse is not turnkey: early sites will still need charging, passenger processing, data connections, and other AAM-specific modifications.⁸

Second, require site-by-site feasibility screening before committing capital. Heliport and rooftop reuse should be treated cautiously because only about 40% of U.S. heliports could accommodate VTOL aircraft with controlling dimensions above 50 feet, and less than 30% could both meet that dimension and comply with other FAA EB 105 cri-

teria. Proposed facilities should also be screened for land use conflicts, nearby residences and schools, obstacle risks, structural capacity, power availability, and multimodal access.⁸

Third, build business cases around full public-sector economics, not just operator revenue. The Orlando Geico Garage case used a \$30 million retrofit assumption, including charging, passenger and crew facilities, MRO, vehicle gates, and structural reinforcement contingency; it also estimated \$600,000 to \$1 million in annual lost parking revenue, potentially offset by a \$2.1 million annual ground lease. Economic-impact claims should be separated from fiscal risk, since the same case estimated 202 operating-phase jobs and about \$35.9 million in value added across Orlando, Orange County, and Florida.⁴

Fourth, use conservative aircraft-performance assumptions in planning. NDARC and SUAVE produced similar gross takeoff weights but materially different component and drag estimates, including nearly 20% higher cruise lift-to-drag in NDARC and a 22% total cruise-drag difference between tools. Battery and motor systems together represented roughly 40% of gross takeoff weight, making hover, climb, wiring, torque, and electric-system assumptions central to feasibility.⁷

Finally, select pilots where infrastructure access matches real demand. Existing TOLA infrastructure in Los Angeles is concentrated in central business districts, leaving surrounding communities underserved and limiting commuter or point-to-point demand outside specific corridors. The next step is therefore a small portfolio of corridor pilots: airport-to-downtown, regional airport links, medical/logistics routes, and rural-connectivity demonstrations, each with explicit go/no-go thresholds for safety, cost, community acceptance, charging readiness, and public benefit.⁶

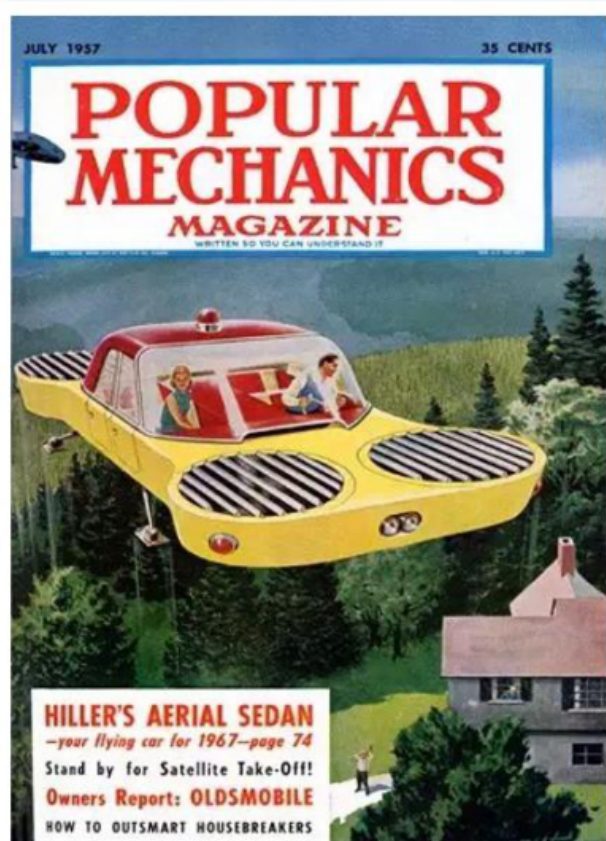


Exhibit. What we now consider to be UAM entered the popular imagination under the guise of “flying cars” as early as the 1950s. For example, a 1957 edition of Popular Mechanics predicted the emergence of flying cars in “the next. Source: S0075.

Study Framing and Methodology

Motivation, scope, and research questions

This paper frames AAM/UAM as a transportation-system feasibility problem, not simply an aircraft-development problem. AAM refers to moving people or property by air between two points in the National Airspace System using advanced aircraft technologies, including electric and eVTOL aircraft, in controlled or uncontrolled airspace. UAM is

the metropolitan subset of that vision, covering passenger and cargo services above populated areas, from package delivery drones to passenger air taxis. ODM is closely related: NASA describes UAM as the subset of ODM focused on metropolitan operations by small passenger- or goods-carrying aircraft over trips of roughly 100 nautical miles or less, while ASSURE treats ODM and UAM interchangeably where automated, electric aircraft provide high-speed on-demand service to the public.¹⁹¹⁰

The motivation is therefore two-sided. On one side, distributed electric propulsion and service-oriented aviation concepts have matured enough to make quiet, efficient, crewed and uncrewed UAM operations technically plausible across missions such as emergency medical evacuation, rescue, humanitarian response, news gathering, traffic assessment, weather monitoring, package delivery, and passenger transport. On the other side, mature UAM requires higher-tempo and higher-density operations than the current ATC system can accommodate without solving technical challenges in airspace integration. Historical experience also counsels caution: commercial UAM-like helicopter services operated in Los Angeles from 1947 to 1971 and in New York from 1949 to 1979, but both ceased after fatal mechanical-failure incidents and financial or operational consequences.⁹

The scope of this research follows that systems framing. A prior ODM thesis defined its purpose as assessing metropolitan ODM networks across technological maturity, regulation, market opportunity, airspace management, and externalities, while explicitly avoiding vehicle development and remaining as technology- and vehicle-agnostic as possible. It focused on opportunities and challenges in the United States, using Southern California as an early-adopter case study while recognizing that feasibility can vary widely across cities and countries. This paper adopts the same practical boundary: feasibility depends on whether vehicles, vertiports, charging, airspace, operating rules, demand, economics, and community impacts can work as one deployable system.⁶

Three research questions organize the analysis:

1. What technological, operational, regulatory, business, and system-interface factors could constrain or prevent AAM/UAM implementation in the United States?⁶
2. What externalities could arise from AAM/UAM network proliferation, what impacts could they have on society, and how could those impacts feed back into implementation feasibility?⁶
3. What near- and long-term technology or policy options could mitigate binding constraints and negative externalities?⁶

These questions require a concept-of-operations lens. A ConOps describes how a system will be used to meet stakeholder expectations, supports requirements and architecture development, and provides a basis for nominal and contingency planning and validation criteria. For AAM/UAM, that means feasibility must be evaluated at the mission, network, infrastructure, regulatory, and social levels at once. Prior ODM research found that constraint severity varies sharply with network scale, so factors that are manageable in near-term pilots may become binding as aircraft volumes grow. Noise, takeoff-and-landing infrastructure, and ATC/airspace throughput illustrate that progression: aircraft noise was identified as the most severe near-term U.S. constraint, TOLA availability as a major near-term infrastructure constraint, and airspace throughput as the most significant far-term scalability constraint.¹⁶

Literature review and data overview

The evidence base combines market forecasts, commercial ecosystem scans, mission-level studies, and localized implementation cases. Market sources define UAM across recurring segments: air taxis, air metro, air ambulance, last-mile delivery, intercity and intracity range classes, piloted/autonomous/hybrid operation, and end users such as ridesharing companies, scheduled operators, e-commerce firms, hospitals, medical agencies, and private operators. A broader taxonomy adds components, platform architecture, propulsion type, maximum takeoff weight, infrastructure, and geography, with vehicle types including air ambulance, air shuttle, cargo air vehicles, personal air vehicles, and last-mile delivery vehicles.¹¹¹²

The market literature is directionally consistent on growth but not numerically uniform. Fortune Business Insights reports the global UAM market at USD 6.02 billion in 2025 and USD 17.53 billion by 2034, with a 2026-2034 CAGR of 14.29%. Roots Analysis reports USD 4.79 billion in 2025, USD 6.63 billion in 2026, and USD 206.79 billion by 2040, with a 27.85% CAGR from 2026 to 2040. These differences make the forecasts useful for bounding opportunity, not as precise demand inputs.¹¹¹²

Across sources, demand is linked to urban congestion, faster travel needs, electric propulsion, and integrated mobility platforms. The same literature identifies the main deployment constraints as certification complexity, evolving regulation, infrastructure gaps, airspace integration, traffic management complexity, battery limitations, and high initial deployment costs. MarketsandMarkets describes its estimation process as a combination of secondary research on eVTOL platforms, infrastructure, airspace management, operating models, and end-user adoption; primary interviews with OEMs, infrastructure providers, operators, regulators, and technology experts; segmentation by system, infrastructure, application, operation mode, range, end user, and architecture; and data triangulation for segment validation.¹³

The reviewed data also supports separating market size from operational feasibility. For example, a Malibu-to-Century City reference mission compares water, direct, and increased-speed routes ranging from 23.7 to 28.8 miles and 7.22 to 7.77 minutes, illustrating how routing and speed assumptions shape service performance. Local implementation data adds another layer: Orlando's Geico Garage analysis used stakeholder interviews, manufacturer/operator public information, IMPLAN modeling, and assumptions on capital cost, operating cost, revenue, employment, and compensation to evaluate a conceptual vertiport retrofit. Together, the literature supports a mixed-method approach: use market reports to define segments and scenarios, mission profiles to test route economics, and site-level studies to evaluate infrastructure, cost, and public-sector exposure.⁶⁴

Workshops and stakeholder engagement

Stakeholder engagement should be treated as a core feasibility method for AAM, because market categories alone do not resolve implementation questions such as site readiness, operating assumptions, public-sector costs, infrastructure conversion, or commercial viability. Market studies commonly validate secondary research through interviews with aircraft OEMs, infrastructure providers, mobility operators, regulators, technology experts, executives, engineers, vertiport developers, and battery and propulsion providers. Those interviews are used to test market-size estimates, operational requirements, adoption trends, infrastructure readiness, airspace integration, certification progress, deployment strategy, and commercial viability.¹³

The Orlando AAM Transportation Plan illustrates how this engagement can be operationalized. For the Geico Garage vertiport concept, VHB interviewed Lilium, reviewed public information from other AAM manufacturers and operators, and used those inputs to set assumptions for capital costs, operating costs, and revenue projections. These assumptions were then presented at the NASA AAM Community Integration Workshop on December 15, 2021, before VHB worked with the Orlando Economic Partnership to model economic impacts using IMPLAN. A separate operator engagement workshop included Lilium, VHB, Metric, and Jacobs, with the objective of discussing data and inputs for the Geico Garage Economic Impact Study.⁴

The same example shows why local public agencies must be involved early. Orlando's parking staff estimated that converting the top deck of Geico Garage could displace roughly \$600,000 to \$1 million in annual parking revenue, based on 100 events per year, 300 repurposed roof-deck spaces, and \$20-\$30 per space. The existing Orlando Magic parking lease would also need renegotiation because it involved 500 spaces per game. At the same time, the analysis suggested a potential \$2.1 million annual ground lease from an AAM operator, though that lease was not included as an IMPLAN input.⁴

This pattern points to a practical engagement model: use industry interviews to bound vehicle, infrastructure, cost, and operating assumptions; use public-agency workshops to test site constraints, foregone revenue, lease structures, and permitting issues; and use regional economic partners to translate those assumptions into economic-impact inputs. In AAM, stakeholder engagement is therefore not just outreach. It is the process that turns speculative market opportunity into auditable local implementation assumptions.

Business case development methodology

Business case development should begin with route-specific economics, not aggregate market forecasts. AAM demand forecasts vary widely because studies use different scopes, socioeconomic assumptions, and deployment constraints; the same literature review cites 2035 passenger-market estimates ranging from \$641 million globally to \$32 billion globally, while another estimate places the U.S. AAM market at \$115 billion by 2035. The practical implication is that market sizing should bound opportunity, while investment decisions should be tested against specific routes, facilities, aircraft assumptions, and operating models.¹⁴

A defensible methodology combines top-down and bottom-up forecasting. The top-down side can use population projections and trip rates from transportation network company operations; the bottom-up side can use GIS to identify candidate city pairs and calculate Haversine distances across selected origins and destinations. This route-level approach is especially important because bottom-up methods require more granular origin-destination data and computational evaluation, but they are better suited to identifying specific corridors than broad national demand estimates. Distance bands should also be explicit, since one hybrid study ranked U.S. routes by trip-miles within eVTOL range-roadmap constraints.¹⁴

The passenger business case should then filter candidate routes through mode-shift logic. The Uber Elevate model envisioned door-to-door journeys combining ground and air passenger modes, with vertiports potentially located at small airports, heliports, parking-garage rooftops, shopping malls, floating barges, roadside facilities, and highway interchange areas. Literature summarized in the route-forecasting study suggests air taxi competition becomes more plausible for longer ground trips: one study found trips above 10 km could compete with air taxis, another found 15-25 km was needed for time savings, and a NASA-commissioned study found that 98% of air taxi demand would come from ground trips exceeding 30 minutes.¹⁴

Facility economics should be modeled as a separate layer. In Orlando's Geico Garage case, analysts used stakeholder interviews with Lilium, public manufacturer/operator information, and assumptions for capital costs, operating costs, and revenue projections, then tested economic impact with IMPLAN. The same case assumed a \$30 million retrofit including charging infrastructure, passenger and crew facilities, maintenance repair and overhaul facilities, vehicle gates, and structural-reinforcement contingency. It also treated opportunity cost explicitly, estimating \$600,000 to \$1 million in lost annual parking revenue and identifying a potential \$2.1 million annual ground lease that was not included in the IMPLAN inputs.⁴

Vehicle and operating-cost assumptions need sensitivity treatment because conceptual eVTOL models remain uncertain. NDARC and SUAVE produced gross takeoff weights differing by about 7%, while major component weights differed substantially, including a SUAVE landing-gear estimate that was 100 lb lighter than NDARC's and judged overoptimistic. Aerodynamic assumptions also diverged: NDARC estimated cruise lift-to-drag nearly 20% higher than SUAVE, and total cruise drag differed by 22%. Because battery and motor weight together represented about 40% of gross takeoff weight, hover, takeoff, climb, torque, wiring, and electric-system efficiency should be carried as business-case variables rather than fixed constants.⁷

Powertrain and autonomy assumptions directly affect operating economics. One eVTOL cost-optimization study used cost per mission as the objective function and found pilot cost and battery cost to be the two most important costs; it identified autonomy, lower battery manufacturing cost, and longer battery life cycle as key levers for reducing mission cost. In a 30 km Los Angeles case, the optimized cost was \$165 per mission, and powertrain and trajectory optimization reduced total direct operating cost by up to 6.6%. Business cases should therefore report base, conservative, and optimistic scenarios for pilot staffing, battery replacement, aircraft utilization, charging turnaround, and route length rather than presenting a single deterministic fare or margin.¹⁵

Econometric modeling, optimization, and sensitivity analysis

AAM economics should be modeled in layers: regional economic impact, facility-level pro forma, network usefulness, and aircraft mission optimization. WSDOT's aviation impact framework separates direct impacts from multiplier impacts, including supplier sales and income re-spending, and measures outcomes through jobs, labor income, value added, and business revenues. Orlando's Geico Garage analysis applies this logic at project scale, using stakeholder interviews, public manufacturer/operator information, passenger revenue, capital cost, employment, and compensation assumptions as IMPLAN inputs.⁸⁴

The econometric layer should distinguish one-time construction effects from recurring operations. In Orlando, IMPLAN was used to estimate direct, indirect, and induced effects on GDP, employment, and taxes, with construction and operations analyzed separately and multi-region input-output analysis used to avoid double-counting City of Orlando, Orange County, and Florida impacts. The case assumed \$30 million in retrofit CapEx, excluded eVTOL vehicle CapEx because aircraft were to be manufactured in Germany, and treated first-year operating output as approximately \$52 million, equal to annual passenger revenue estimates. Reported operating impacts totaled 202 jobs, \$35.9 million in value added, and 20-year fiscal impacts of \$4.4 million for the City of Orlando, \$14.8 million for Orange County, \$45.2 million for Florida, and \$38.0 million federally.⁴

Facility models also need sensitivity around opportunity cost and ownership structure. Orlando's base case included charging infrastructure, passenger and crew facilities, MRO, vehicle gates, and structural-reinforcement contingency, but noted that a detailed structural assessment was outside scope. It also identified \$600,000 to \$1 million in annual lost parking revenue and a possible \$2.1 million annual ground lease, while excluding that lease from IMPLAN inputs. These exclusions materially affect public-sector affordability even when regional GDP and tax impacts look favorable.⁴

Optimization should connect business assumptions to aircraft physics. One eVTOL cost-optimization study minimizes cost per mission through integrated powertrain, mission, mass, and cost models, arguing that powertrain efficiency and power density alone are insufficient because powertrain choices affect reliability, cost, and energy distribution across mission segments. Its optimization variables include switching frequency, DC bus voltage, rated output power, winding diameter, converter topology, battery mass from 200 to 800 kg, and maximum range from 30 to 100 km.¹⁵

Sensitivity analysis should therefore report ranges, not point estimates. As cruise mission range increases, the powertrain has greater effects on direct operating cost, gross aircraft mass, battery mass, powertrain mass, electricity cost, and battery depreciation; longer range also increases current, wire thickness, cooling needs, and conduction losses once voltage reaches its preset maximum. Higher battery energy density reduces direct operating cost and gross mass, but at diminishing rates, shifting dominant cost contributors toward airframe, maintenance staff, and pilot salaries; battery life cycle mainly affects battery cost, while switching frequency creates a tradeoff between powertrain mass and battery mass.¹⁵

Market Context and Use Cases

On-demand mobility and aviation

On-demand mobility aviation should be treated as a service-network problem before it is treated as an aircraft problem. Candidate operations span inter-urban flights between nearby cities and intra-urban flights within a single metropolitan region; commuter airline analogues typically use 6-9 passenger fixed-wing aircraft on routes under 100 miles, while ODM networks and helicopter charters are associated with metropolitan intra-urban service.⁶

The strongest early learning environments may not be mass-market passenger networks. Novel aircraft and automation are likely to gain flight experience first in general aviation because barriers to entry are lower than for commercial Part 135 operations, while public-service markets such as EMS, law enforcement, and news gathering already have decades of helicopter operating experience and may adopt ODM aircraft before passenger networks become viable. This matters because commercially scalable ODM depends on more than vehicle certification: operators must also learn dispatch, route design, airspace access, turnaround, maintenance, and public acceptance under real operating constraints.⁶

Los Angeles illustrates why route context dominates generic market logic. The basin was selected in one ODM case study because of its large consumer base, existing helipad infrastructure, mild weather, polycentric structure, severe roadway congestion, and presence of helicopter charter operators. The same study defined the case boundary using commuter flows, mission characteristics and aircraft range, current helicopter charter services, and Southern California population density. Long-distance commuter flows into Los Angeles suggested potential inter-community ODM missions, while dense flows around the Los Angeles central business district suggested short intra-city demand with different vehicle requirements and flight profiles.⁶

The operational concept is inherently multimodal. Integrated UAM business models typically combine ground access to a vertiport, an eVTOL flight, and a final ride-hailing segment to the destination. Passenger drones are described as most efficient for distances exceeding 20 km, and their value proposition depends on swiftness, on-demand availability, pooling, and integration with broader mobility platforms. This makes total journey time, not airborne time alone, the decisive service metric: one cited concern is whether air-taxi time savings meet customer expectations across the entire door-to-door journey.¹⁶

Business models remain unsettled. AAM vehicle businesses are commonly framed around the degree of vertical integration, from pure vehicle integration and pure vehicle operation to exclusive or non-exclusive vertically integrated operators. Some manufacturers have considered building and operating their own takeoff-and-landing infrastructure, although interviews indicate some are moving away from this model because of capital requirements. Passen-

ger services may resemble scheduled regional aviation, high-net-worth charter, rideshare shuttle service, or app-based multimodal on-demand travel with pricing tied to route, timing, and network demand. Cargo models may be business-to-business logistics services or internal fleets where consumers pay directly for delivery.¹⁷

Mission flight profiles show why “direct route” is rarely a complete answer. For a Malibu-to-Century City reference mission, a water route maximized over-water flight to reduce ground noise exposure, operated in Class G and E airspace, required Santa Monica Class D access during certain hours, and was limited to 200 kt. The same mission still required ATC permission between 7:00 a.m. and 9:00 p.m. for all three studied profiles because Century City lies within Santa Monica surface-level controlled airspace. Alternative profiles could climb to 5,000 ft MSL and enter overhead Class B airspace, enabling flight at up to 250 kt but requiring LAX ATC approval and routing.⁶

Longer missions compound these tradeoffs. For San Diego-to-Los Angeles, four profiles were developed: water, direct, inland, and coastal. The water route reduced ground noise by maximizing ocean overflight and avoided about 14 nautical miles over dense areas, but required Los Angeles Class B clearance. The direct route was close to line-of-sight but passed near multiple airspace constraints and helicopter routes, requiring congestion consideration near the Long Beach Freeway helicopter VFR route and nearby helicopter corridors. The inland route avoided open water and minimized dense-population overflight by following mountain terrain, but it was the longest route and encountered VFR and helicopter operations near several airspaces. The coastal route avoided open water yet passed over populated areas at altitude and could require rerouting or higher flight if restricted Camp Pendleton airspace was active.⁶

The key operational risks are therefore structural, not incidental. ODM scale could substantially increase low-altitude ATC workload, and controllers may be unable or unwilling to admit all operations into controlled airspace under current procedures. Higher vehicle density raises separation challenges among ODM aircraft, general aviation, commercial aircraft, and low-altitude UAS. Frequent community and business-location operations also intensify noise exposure, an issue already significant enough in Los Angeles aviation to draw U.S. congressional attention.⁶

Automation may improve operating economics, but it adds certification and assurance burdens. Current work on aviation machine learning emphasizes trustworthiness analysis, assurance, human factors, and safety-risk mitigation, while recognizing that no single methodology is sufficient for safe integration. UAM navigation also faces urban accuracy, availability, and integrity challenges, especially around vertiports, and standalone GNSS may be insufficient in dense urban areas. In practice, on-demand aviation becomes feasible only when demand, multimodal access, route geometry, airspace permission, aircraft performance, turnaround, noise, and safety assurance work as one operating system.¹⁸

AAM/UAM market opportunity and current status

AAM is no longer only an aircraft concept; it is an emerging transportation ecosystem spanning vehicles, operators, maintenance, vertiports, charging, airspace management, certification, and public acceptance. One glossary defines AAM as air transport of people or property between two points in the NAS using advanced aircraft, including electric aircraft and eVTOLs, in both controlled and uncontrolled airspace; UAM is the metropolitan subset covering passengers and cargo, from package drones to passenger air taxis above populated areas. This framing matters because market readiness depends on the whole operating system, not only on vehicle performance.¹

Commercial interest is substantial. A 2026-2036 market report describes eVTOL and AAM as a convergence of aerospace engineering, electric propulsion, battery technology, autonomy, and digital infrastructure, with investment from aerospace, automotive, technology, and sovereign-capital actors. The same source frames the sector through a “5As” ecosystem: aircraft, ancillary services such as MRO, airlines/operators, airports/vertiports, and airspace/traffic management. This supports a broader opportunity thesis: value may accrue not only to OEMs, but also to infrastructure developers, battery and propulsion suppliers, operators, training providers, and certification or airspace-service firms.³

The aircraft landscape is still sorting into architectures with different operating implications. The reported principal eVTOL configurations are multicopter, lift+cruise, tiltwing, and tiltrotor; multicopters emphasize simplicity for short urban trips, lift+cruise separates vertical and forward-flight systems for better cruise efficiency, and vectored-thrust designs offer higher range and speed at greater complexity. Market analysis now extends beyond small urban air taxis into regional travel, logistics, emergency response, cargo delivery, air ambulance, rural rideshare, sub-regional shuttle, and urban private-hire use cases. The strongest cited distance band is not the shortest urban hop: eVTOL

economics are described as most compelling at roughly 40-160 km, where congestion reduces the speed advantage of surface transport.³

Forecasts remain ambitious but should be treated as scenario inputs rather than proof of inevitability. One market source projects passenger UAM revenue rising from roughly US\$1 billion around 2030 to US\$90 billion annually by 2050, with 160,000 commercial passenger drones operating globally. The same source reports eVTOL startup funding rising from US\$40 million in 2016 to US\$907 million in the first half of 2020, and exceeding US\$6.5 billion in 2025. Yet it also identifies certification and regulation as the “single greatest determinant” of market timing and notes that type certification has been more costly and time-consuming than projected, pushing commercialization targets later.³

Current status is therefore mixed: technology demonstration, capital formation, and planning activity are real, but deployment remains constrained. Battery technology is identified as the foremost technical challenge, with current lithium-ion cells at 250-300 Wh/kg while commercially viable operations are said to require 400-500+ Wh/kg. Public perception, noise, social license, vertiport infrastructure, air traffic management, and UTM/ATM integration are all explicit components of current market analysis. The sector is also moving unevenly by region: North America is described as leading in OEM development and regulatory progress, Europe benefits from EASA's framework, China is developing through low-altitude-economy policy, and the Middle East is investing through smart-city strategies.³¹⁹

Public agencies are beginning to translate the opportunity into planning programs. PennDOT's AAM and Airport Electrification Feasibility Study evaluated use cases, infrastructure needs, policy and regulatory issues, funding opportunities, and airport electrification as an enabling component of deployment. Its literature review covered more than 50 state, federal, industry, TRB, ACRP, and academic sources, and identified regulatory alignment, vertiport placement, grid capacity, multimodal connectivity, public perception, and community engagement as central issues. This is the practical state of the market: large opportunity, fragmented readiness, and a near-term need for disciplined route, infrastructure, certification, and community validation before scale claims can be trusted.²⁰

Last-mile delivery, air metro, and air taxi

NASA's 2018 UAM market study, conducted by Crown Consulting, separated near-term urban applications into three use cases: last-mile delivery, air metro, and air taxi across 15 U.S. cities. Last-mile delivery is defined as unscheduled real-time package delivery from local hubs to receiving vessels, with routes determined as online orders are placed. The NASA market assumptions describe this use case as small UAS carrying roughly 5 lb over about 10 miles roundtrip, supported by receiving vessels, distribution hubs, docking and charging stations, and UTM.¹⁶²¹

Air metro is the more transit-like concept: predetermined routes, regular schedules, and fixed stops in high-traffic areas, using 2- to 5-passenger autonomous VTOL aircraft over roughly 10-70 miles per trip. Its infrastructure assumption is substantial but structured: roughly 100-300 vertiports per metropolitan statistical area, with high-traffic sites able to handle about 3-6 VTOLs at once, plus charging, service stations, and UTM. The market model cited in the NASA study estimated air metro in-year profitability in 2028, with 130 million passenger trips, 4.1 thousand vehicles, and a \$50 trip price in the first profitable year, rising by 2030 to 740 million trips, 23 thousand vehicles, and a \$30 trip price, assuming regulatory challenges are overcome.²¹

Air taxi is the most flexible and most infrastructure-intensive of the three. The NASA framing defines it as door-to-door ride-hailing or ride-sharing in which passengers select pickup and destination rooftops, using autonomous 2- to 5-passenger VTOL aircraft over roughly 10-70 miles per trip. Unlike air metro, routes are unscheduled and likely different each time, requiring a very large density of vertistops on or near buildings to approximate door-to-door service. That density is the core economic obstacle: the model assumes walking time of less than three minutes to a stop, making widespread infrastructure across metropolitan areas unlikely by 2030. The NASA study therefore judged widespread air taxis unprofitable under then-current constraints, while allowing possible viability in dense, high-income markets such as Manhattan, Boston, San Francisco, Miami, or Philadelphia.²¹

The regulatory distinction is equally important. The NASA market study stated that scalable UAM operations were not permitted under the then-current regulatory environment: last-mile delivery was heavily restricted and available only through waivers and certificates of authorization, while air metro and air taxi were permitted only as traditional crewed helicopter services, excluding autonomy and eVTOL design from the full business case. Enabling all three use cases would require regulation across traffic and fleet operations management, vehicle development and production, airspace design and implementation, individual vehicle operations, and community integration. The same

source placed scalable last-mile delivery on a nearer-term regulatory path of roughly 2-5 years, but air metro and air taxi on a mid- to long-term path of roughly 10 years or more absent major changes.^{[21](#)}

Public acceptance is not a secondary issue. In the NASA consumer survey, 25% of more than 2,500 consumers reported comfort with unmanned aerial technology, about 25% said they would not use UAS or eVTOL services when widely available, and nearly half were potentially comfortable with delivery and UAM use cases. Concerns clustered around safety, privacy, job security, environmental threats, noise, and visual disruption. For last-mile delivery, respondents emphasized vehicle malfunction, package theft, and camera-related privacy concerns; for passenger UAM, they emphasized passenger and bystander safety and high operating costs. Proven safety records and demonstrations were identified as the factors most likely to increase consumer comfort.^{[21](#)}

Taken together, these use cases imply different entry paths rather than one generic UAM market. Last-mile delivery has lighter payloads and a nearer regulatory horizon but faces privacy, theft, BVLOS, over-people, and UTM constraints. Air metro has a clearer pooled-service logic and lower vertistop-density burden than air taxi, but depends on scheduled routes, high passenger volumes, autonomous VTOLs, and metropolitan vertiport networks. Air taxi offers the strongest on-demand value proposition, yet its economics are most exposed to infrastructure density, low occupancy assumptions, affordability, and the difficulty of matching door-to-door ground mobility.^{[21](#)}

Airport shuttle, regional travel, freight, and medical transport

Airport shuttle, regional travel, freight, and medical transport are best treated as adjacent AAM markets rather than simple extensions of short-hop urban air taxi service. The common advantage is that these use cases can be organized around known nodes, recurring corridors, or high-value time savings, reducing some of the siting and routing complexity that weakens fully flexible air taxi economics. Even so, the evidence base remains immature: VTOLs are still described as a relatively new technology with few evidence-based studies of regular public or personal urban use.^{[22](#)}

Airport shuttle is the clearest bridge between today's aviation system and AAM. It can use fixed origin-destination pairs, predictable peak periods, and airport-adjacent passenger demand, while avoiding the extreme vertistop density required for door-to-door air taxi. The Orlando AAM plan illustrates the infrastructure scale: a conceptual downtown vertiport at Geico Garage was analyzed as part of a 14-vertiport Florida network, with a \$30 million retrofit assumption covering charging infrastructure, passenger and crew facilities, maintenance repair and overhaul facilities, vehicle gates, contingency, and likely structural reinforcement. That case also shows the opportunity cost of urban siting: converting the garage roof could displace an estimated \$600,000 to \$1 million in annual parking revenue, although a potential \$2.1 million annual ground lease to an AAM operator was identified as an offset.^{[4](#)}

Regional and sub-regional travel may be more operationally natural than dense urban roaming because flights can be planned around fewer nodes, longer trip distances, and more stable schedules. UAM network studies describe low-altitude operations up to 2,000 ft, distances up to 100 km, typical flight durations of 10-15 minutes, turnarounds up to 15 minutes, speeds of 80-150 kt, and high-frequency connections among vertiports. Profitability, however, depends on network design: in the São Paulo case summarized in the literature, a fully connected network was not profitable at US\$5.00 per km, while network optimization was identified as critical to improving profitability.^{[23](#)}

Freight and medical transport broaden the value proposition beyond passenger willingness to pay. AAM is described as enabling low-altitude operations for passenger transportation and cargo delivery, and one cited impact assessment evaluates benefits for passengers, patients, farmers, logistics companies and customers, and bridge inspection entities. The design literature also identifies air ambulance cost analysis as an eVTOL application area, indicating that medical use cases are being studied separately from general air taxi markets. These applications may justify deployment where urgency, access, or service reliability matters more than mass-market fare parity, but the provided evidence does not establish commercial readiness.^{[23](#)[24](#)}

Across all four use cases, infrastructure and charging remain binding assumptions. Existing hospital helipads, other helipads, underused parcels, and open spaces may support limited initial VTOL operations, but most cities are not described as having enough takeoff and landing sites for broad VTOL deployment. Freeway cloverleaf interiors have been proposed as vertiport or vertistop locations because ambient freeway noise and public ownership could help with access and acceptability, though some sites would still require car or bus access because walking out of an interchange is impractical. Battery charging is also operationally consequential: one cited eVTOL example required up to 5.5 hours to charge an 8-kWh battery under low-power input conditions, while the study used a one-hour assumption for charging a 100-kWh battery as a more reasonable operating estimate.^{[22](#)}

The environmental case is similarly route- and grid-dependent. VTOL life-cycle analysis must include manufacturing and operations, including energy consumed, environmental externalities, and operating costs. For eVTOLs, operational emissions are tied primarily to electricity generation for battery charging and vertiport energy needs. Texas grid assumptions in one study showed ERCOT CO₂e emissions falling from 239 kg/MWh in 2016 to 223 kg/MWh in 2018 and 195 kg/MWh in 2021, underscoring that eVTOL externalities depend on local power generation over time.²²

The practical implication is that airport shuttle and regional routes should be the first candidates for rigorous business-case testing, while freight and medical transport should be evaluated where operational urgency or social value can support higher costs. None should be treated as automatically scalable. Each requires corridor-level validation of demand, vertiport access, charging turnaround, airspace integration, life-cycle impacts, and facility economics.

Business cases for AAM implementation and airport electrification

AAM business cases should be treated as public-infrastructure investment cases, not only operator revenue cases. PennDOT's feasibility study framed AAM and airport electrification as a readiness, policy, infrastructure, and funding problem, supported by literature review, airport electrification assessment, tailored AAM business cases, hot-spot mapping, project evaluation, and funding review. Its business-case structure explicitly separated AAM implementation from airport electrification, with dedicated analysis for market conditions, electrification use cases, technology gaps, and initiatives.²⁰

For AAM implementation, the strongest public-sector case is enabling infrastructure where private operations may later prove viable. PennDOT's stated priorities emphasize publicly accessible transportation infrastructure, public benefit, equity, emergency services, and underserved communities, while its role is positioned around infrastructure support, public-sector pilot demonstrations, interagency coordination, and grant facilitation rather than initially funding commercial operations directly. This makes the near-term business case less about subsidizing air taxi service and more about reducing deployment friction through shared-use vertiports, pilots, public-private partnerships, and grant-ready project packages.²⁰

The Orlando Geico Garage case illustrates the site-level economics that such business cases must test. The concept assumed a downtown vertiport within a 14-vertiport Florida network, used a \$30 million retrofit assumption, and included electric charging infrastructure, passenger and crew facilities, MRO facilities, vehicle gates, and structural reinforcement contingency. Because the garage was not originally designed for rooftop aviation use, the analysis noted that structural reinforcements would likely be required and that a detailed structural assessment would be needed to understand cost and risk. The same case estimated \$600,000 to \$1 million in annual lost parking revenue from converting rooftop parking, but also identified a potential \$2.1 million annual ground lease to an AAM operator, which could offset that loss. Its IMPLAN analysis estimated 202 total jobs, \$35.9 million in value added, and 10-year tax impacts of \$2.8 million for the City of Orlando, \$9.6 million for Orange County, \$29.3 million for Florida, and \$24.6 million federally.⁴

Airport electrification has a narrower but more immediate infrastructure logic. PennDOT distinguishes AAM funding as broader, covering on- and off-airport infrastructure, vertiports, multimodal hubs, last-mile connectivity, operator incentives, supply-chain development, and nontraditional aviation facilities, while airport electrification funding focuses on airport sustainability, operational efficiency, clean energy, electric ground support equipment, zero-emission vehicles, power systems, and on-airport infrastructure. The funding review examined 85 programs; the AAM funding task identified 73 programs, 34 of which were relevant to airport electrification, and a targeted airport-electrification review added 12 more, yielding 46 applicable electrification opportunities.²⁰

Federal programs are central to the electrification business case. PennDOT identifies FAA AIP, VALE, FAA Reauthorization Act Sections 742, 745, and 960, USDOT SMART, RAISE, TIFIA, EDA Revolving Loan Fund, and FTA Low or No Emission programs as funding pathways that can support airport electrification, smart technology, multimodal sustainability, clean energy, and transportation investments. AIP alone is described as allocating about \$3.2 billion annually for airport capital projects, with more than \$10 billion distributed nationally from 2022 to 2024 and more than \$268 million to Pennsylvania airports; during that period, over \$73 million supported electrification-related projects. VALE has supported 141 projects at 58 airports since 2004, including Pennsylvania investments in pre-conditioned air, ground power, recharging infrastructure, and EVs.²⁰

The practical implication is that airport electrification can proceed before full AAM market maturity. Electrified GSE, gate power, charging, clean-energy systems, and terminal energy upgrades can improve current airport operations

while preserving optionality for future electric aircraft and vertiport demand. AAM implementation, by contrast, remains more contingent on route economics, siting, community acceptance, and regulatory integration; WSDOT notes that AAM will involve new aircraft, infrastructure, operations, markets, and places, functioning as a complementary mode linking aviation to other modes and geographies. Therefore, the most defensible business case sequence is to fund near-term electrification and planning assets with current operational value, while advancing AAM pilots and shared infrastructure where public benefits, emergency use, underserved access, and private operating interest overlap.²⁰⁸

Demand and Reference Missions

Consumer demand estimation and geographic locating

Consumer demand for AAM should be estimated at the trip-chain level, not at the aircraft level. Time savings depend on the full journey: access to a vertiport, waiting and processing time, the air segment, and egress to the final destination; long wait or processing times can erase the speed advantage of direct flight. Recent simulation frameworks therefore model UAM as a multimodal service, assigning travelers across auto and UAM segments while also tracking vertiport capacity, departure separation, passenger travel time, addressable market, operational benefit, and aircraft utilization.¹⁸²⁵

The most defensible demand-screening variables are distance, congestion, access time, income or willingness to pay, and concentration of origins and destinations. Air taxi is expected to be most attractive for long-distance urban trips in congested cities because it can bypass highway congestion, but first-/last-mile and loading/unloading time make shorter trips less attractive. This implies that geographic locating should prioritize corridors where ground travel is both long and unreliable, and where users can be pooled through concentrated pickup or drop-off points.²⁶

The Los Angeles ODM case study used several demand lenses rather than a single market proxy. Its demand-estimation subsection explicitly examined helicopter charter demand, replacement of long commutes, and demand from wealthy commuters. Its supporting data structure included comparisons of mobility-mode performance and cost, helicopter charter pricing, helicopter charter fares per passenger mile, top U.S. mega-commuter flows, and common California mega-commuter routes. This is a useful template because it combines observed premium aviation behavior, surface-transport pain points, and demographic adoption plausibility.⁶

Geographic locating also needs stress tests outside obvious early-adopter corridors. In the L.A. study, three random reference missions were generated to avoid biasing operational conclusions toward author-selected early markets. The random points were selected from all roads within a rectangular region roughly bounded by San Bernardino, Malibu, Irvine, and Antelope Valley, rather than only from valid addresses, because address-only sampling would overrepresent dense home and business clusters and underrepresent suburban or rural areas. That choice matters because lower-density or peripheral travelers may still have high willingness to adopt air service when surface access is poor or trip times are severe.⁶

The resulting random missions show why route-level validation is necessary. San Bernardino to Perris was a 26-mile edge-city trip taking about 28 minutes in low traffic and up to 70 minutes in evening rush hour, a 150% congestion penalty. Arleta to Corona was a 71-mile suburban-to-suburban trip taking about 70 minutes in low traffic and up to 190 minutes in evening rush hour, a 170% congestion penalty. Altadena to Culver City was a 30-mile trip taking about 40 minutes in low traffic and up to 130 minutes during morning rush hour, a 225% congestion penalty. These cases suggest that demand geography is not simply “downtown to airport”; it can emerge from edge-city, suburb-to-suburb, and congestion-crossing patterns.⁶

The Antelope Valley to Los Angeles case adds a different demand pattern: persistent distance burden rather than only peak congestion. The study found that this route required over 75 minutes by automobile each way regardless of rush hour, and identified 951 individuals commuting from the Lancaster reference region to downtown Los Angeles. Because the destination was highly concentrated in downtown Los Angeles, the study argued that ODM operations could benefit from passenger pooling and centralized pickup/drop-off.⁶

Consumer acceptance remains a demand constraint, not a separate public-relations issue. NASA-commissioned work summarized in the literature found that almost half of surveyed consumers were open to UAM, but safety, privacy, job security, environmental effects, noise, visual pollution, passenger and bystander safety, and operating cost were major concerns. Demonstrated safety records and practical demonstrations were identified as ways to in-

crease consumer confidence. Thus, geographic demand maps should be treated as conditional: they identify where AAM could be useful, but adoption still depends on safety credibility, access convenience, service reliability, and total trip economics.¹⁶

Long-commute, helicopter charter, and wealthy commuter demand

Early AAM demand is best treated as a set of narrow adopter contexts, not a mass-market assumption. Vascik's Los Angeles ODM analysis starts from a mode-choice premise: travelers choose among available mobility modes based on preferences that vary by person and context, so demand should be screened where speed has unusually high value despite higher operating cost. The strongest candidate contexts are long-distance trips, heavy-congestion trips, event or popular-destination trips, airport or hospital trips with delay consequences, and trips by high-wealth consumers with greater willingness to pay.⁶

The Los Angeles case therefore used three empirical demand lenses: existing helicopter charter services, census evidence on exceptionally long commutes, and wealth proxies based on home valuation and average household income. This is a useful structure because it separates proven aviation willingness-to-pay from ground-trip pain and from income-based affordability. It also avoids assuming that all congested travelers are addressable: private cars still dominated the Southern California commuter portfolio, representing more than three fourths of commuter trips, while helicopter and aircraft charter were not significant mobility shares.⁶

Current helicopter charter demand is a small but important revealed-preference market. Helicopter and aircraft charter services were limited partly because they were too expensive for most travelers and, outside specific routes, did not provide enough availability or geographic coverage to justify the price premium even for consumers who could afford them. The strategic AAM opportunity is therefore not to copy legacy charter at the same cost structure, but to occupy the lower-cost end of today's charter market while also attracting high-cost, long-distance automobile trips if ODM networks can provide broader availability, better coverage, and lower cost than traditional charters.⁶

Long-commute demand is geographically plausible in Southern California because Los Angeles County draws workers from a large commuter basin, including distant inland communities, and these long-distance inter-community flows were identified as potential ODM missions. Dense commuter flows around the Los Angeles CBD also suggested a separate intra-city demand layer with different vehicle requirements and flight profiles from longer inter-community routes. Interconnections among CBDs and nearby commuter basins further supported expanding the system boundary to include Santa Barbara and San Diego.⁶

Wealthy commuter demand is analytically distinct from long-commute demand. The Los Angeles study treated home value and average household income as proxies for communities with higher willingness to pay, while separately using commute duration and distance to identify travelers likely to value time savings. This distinction matters because high income alone does not create a compelling air trip; the origin-destination pattern must still align with infrastructure, access time, route geometry, and congestion relief.⁶

The reference missions derived from these demand screens show why early markets may be corridor-specific. The study used current helicopter charter routes, mega-commuter routes, and high-income commuter routes as the foundation for nine likely early-adopter reference missions, then added three random missions to expose challenges not captured by obvious promising corridors. Non-commute point-to-point demand was expected to be less temporally and directionally consistent than commuting demand, with business, recreation, and private trips distributed more sporadically through the day.⁶

Several non-commute examples overlap with charter-style and wealthy-traveler demand. Inter-city missions include major city-pair travel such as San Diego to Los Angeles, where helicopter charter companies already offer service; that reference mission was 122 miles, with driving time estimated at 110 minutes without congestion and up to 240 minutes during peak periods. Shorter inter-city missions also matter in polycentric Los Angeles: the region was described as having 45 circumferential employment sub-centers, supporting potential demand between the CBD and edge cities or among edge cities during peak congestion. Recreational missions such as San Marino to Palm Springs were tied to long-distance travel through congested ground networks and geographic barriers.⁶

The demand implication is narrow but actionable: early passenger AAM is most defensible where multiple filters overlap, such as high willingness to pay, long or unreliable ground travel, existing premium aviation behavior, concentrated origins and destinations, and feasible landing infrastructure. Broader adoption remains constrained by the

same issues that limited helicopters. NASA's passenger-acceptance work notes that helicopters have historically served urban premium transport, but cost and noise severely limited their ability to scale, with economic success concentrated in price-insensitive niches such as emergency medical evacuation and high-value passenger transport. UAM market viability similarly depends on large demand and passengers' willingness to pay a premium for time savings and reliability over ground alternatives.^{[27](#)}

Passenger trips and goods trips

Passenger AAM demand is best interpreted by trip type rather than by a generic "urban air taxi" label. In the Los Angeles ODM case study, non-commute point-to-point missions were defined because business, recreation, tourism, private transportation, airport access, event travel, and medical trips do not follow the same temporal or directional pattern as commuting; unlike commute demand, they may be sporadic across the day and dispersed across many origin-destination pairs. This matters for fleet planning: passenger demand may be high-value without being smooth, predictable, or bidirectional.^{[6](#)}

The strongest passenger cases combine distance, congestion, and destination value. The San Diego City Center to Los Angeles City Center reference mission was 122 miles, with driving time estimated at 110 minutes without congestion and up to 240 minutes at peak periods, a 118% congestion penalty. A later travel-time analysis showed moderate bimodal congestion from roughly 5:00-8:00 AM and 2:00-5:00 PM, but also suggested that the long distance alone could support ODM demand beyond rush-hour windows. By contrast, short intra-metropolitan missions depend more heavily on peak congestion, deadlines, or special-purpose demand, as illustrated by airport, stadium, and hospital reference missions in the Los Angeles study.^{[6](#)}

Passenger economics are distance-sensitive. One road-versus-air study found that electric air travel could match or outperform gasoline road travel for passenger trips, consolidated goods trips, and unconsolidated goods trips when airspace is uncongested. For a 50-mile passenger movement, road travel took 42 minutes longer than air in uncongested conditions and 115 minutes longer in congested conditions; the same comparison estimated fuel-cost savings of \$2.41 and \$5.79, and CO₂ reductions of 25.13 and 47.17 pounds, respectively. The same source cautioned that road remained cheaper for trips of 20 miles or less in uncongested conditions and 10 miles or less in congested conditions, and road also produced less CO₂ for uncongested trips under 10 miles and congested trips under 5 miles.^{[28](#)}

Goods movement has a different feasibility threshold. Consolidated goods movement by small aircraft was found to be competitive with diesel delivery trucks only if aircraft energy consumption stayed below 25 Wh/mile. For unconsolidated goods movement, the same study found that a small drone was about 350 times better than an average sedan in both fuel cost and emissions. This implies that goods use cases may scale first where payloads are small, routing is direct, and the comparison mode is inefficient point delivery rather than full truckload logistics.^{[28](#)}

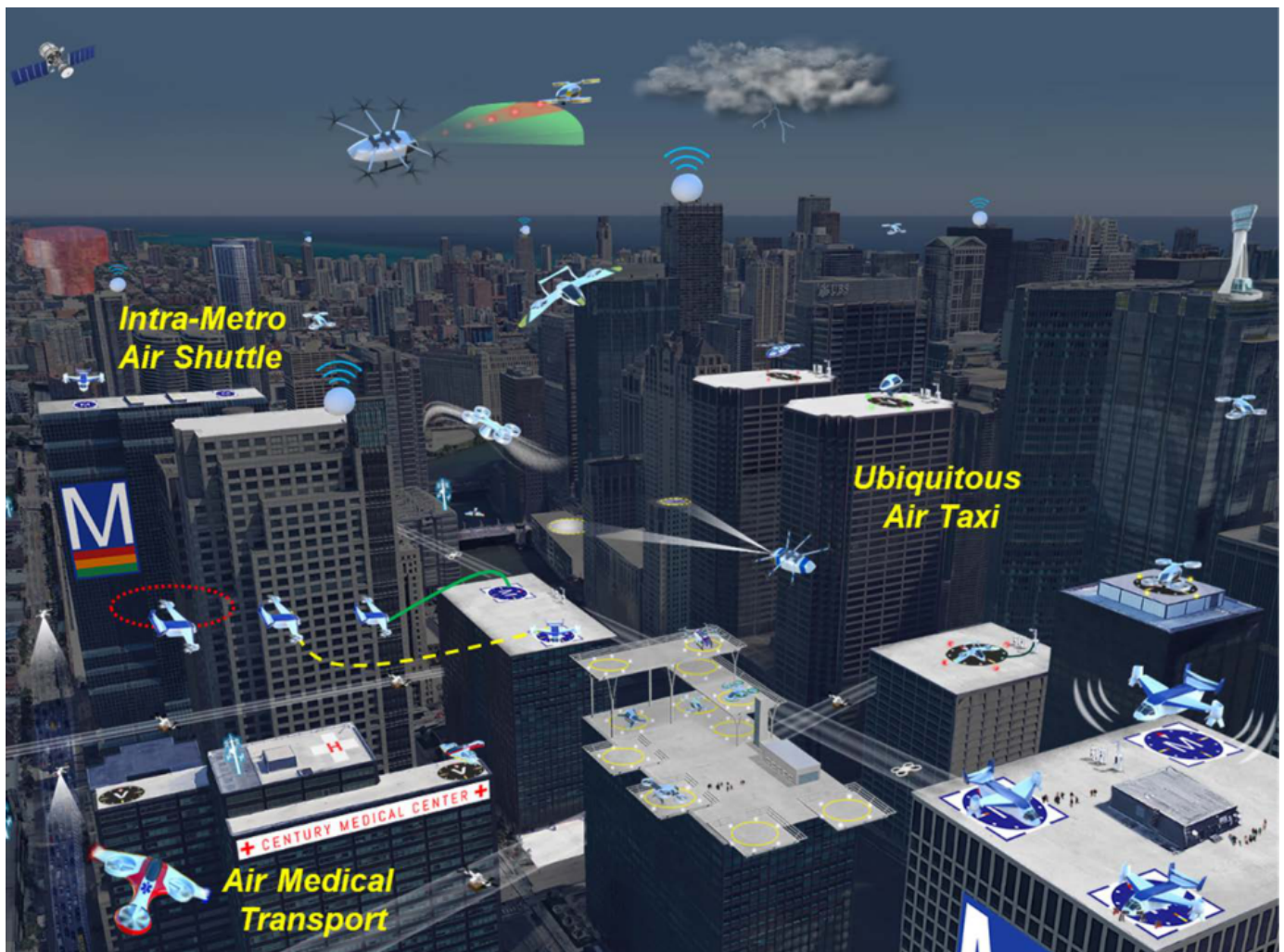


Exhibit. Figure 10 illustrates conceptual air vehicles performing three reference missions in the vision for the mature state. Source: S0039.

Los Angeles case study system boundary

The Los Angeles case study used the L.A. basin and greater Southern California as its initial ODM Aviation boundary because the region combined several early-adopter traits: a large consumer base, polycentric urban form, severe roadway congestion, existing helipad infrastructure, mild weather, and active helicopter charter operations. This made Los Angeles useful not as a generic “flying car” market, but as a stress test for whether demand, infrastructure, weather, operations, and community constraints could align in one metropolitan system.⁶

The boundary was defined through four screening factors: commuter transportation flows into metropolitan Los Angeles, expected ODM mission characteristics and aircraft range, current helicopter charter services, and population density across Southern California communities. Commuter-flow mapping from the 2006-2010 American Community Survey showed that Los Angeles County drew workers from a large part of Southern California, including inland communities at substantial distance, making long-distance inter-community trips relevant to the case study. At the same time, dense commuter flows around the Los Angeles CBD indicated potential short-distance intra-city demand, so the study treated the boundary as layered rather than uniform.⁶

This layered structure matters because short intra-city operations and longer inter-community operations impose different aircraft and network requirements. The study therefore considered an inner Los Angeles boundary for dense urban trips while also expanding the broader system boundary to include connected Southern California markets, including Santa Barbara and San Diego CBDs, because commuter-basin interconnections suggested relatively stable route demand between nearby centers. The “megaregion” concept supported this expansion: Southern California cities are linked not only by geography, but also by infrastructure, economic relationships, environmental systems, topography, culture, history, and land-use patterns.⁶

The boundary was also tied to mission feasibility, not only demand density. The thesis distinguished inter-urban flights between nearby cities from intra-urban flights within one metropolitan region, and noted that commuter air-

line routes are typically below 100 miles while some short routes can be around 20 miles. That framing allowed the Los Angeles case to include both urban air taxi-like operations and longer thin-haul or regional missions within one reference geography.⁶

The boundary then fed directly into reference-mission design. The broader methodology identified early adopting markets using commuting, income, household valuation, and current helicopter charter service information, then defined twelve reference missions to span mission range, demand, ground population density, infrastructure availability, and airspace interaction. Those missions were evaluated through technical feasibility, regulatory compliance, community acceptance, air traffic control interfaces, ground infrastructure, surface-transportation interfaces, and environmental impact. In this sense, the Los Angeles system boundary was not a map outline alone; it was the operational envelope within which business case, vehicle-agnostic ConOps, infrastructure needs, airspace interactions, and public constraints could be compared.⁶

Reference mission definitions and mission flight profiles

Reference missions translate the Los Angeles system boundary into testable operating cases. Vascik's case study organized missions into three groups: daily commute, non-commute point-to-point, and randomly selected missions, then summarized the full set before moving into ConOps evaluation. This structure matters because ODM feasibility changes materially with trip purpose, geography, infrastructure access, and airspace interaction; a commuter corridor, an airport transfer, a hospital trip, and a randomly selected OD pair stress different parts of the system.⁶

The daily-commute set included Malibu-Century City, San Bernardino-Glendale City Center, and Antelope Valley-Los Angeles City Center. The non-commute set added San Diego-Los Angeles City Center, Los Angeles City Center-Long Beach, Beverly Hills Hotel-LAX, Redondo Beach-Dodger Stadium, Rancho Palos Verdes-Torrance Memorial Hospital, and San Marino-Palm Springs. The randomized set covered San Bernardino-Perris, Arleta-Corona, and Altadena-Culver City. Supporting tables were prepared for mission specifications and ground-transportation comparisons across these routes, including travel-time and average-speed figures for San Bernardino-Glendale, Antelope Valley-Los Angeles, San Diego-Los Angeles, Los Angeles-Long Beach, Beverly Hills-LAX, Redondo Beach-Dodger Stadium, Rancho Palos Verdes-Torrance Memorial Hospital, San Marino-Palm Springs, San Bernardino-Perris, Arleta-Corona, and Altadena-Culver City.⁶

The flight profile is not just the airborne segment. NASA's UAM market study modeled air taxi and airport shuttle trips as a chain of request, ground wait, ground access to heliport or vertiport, air-side wait and embarkation, climb, cruise, descent, disembarkation, and ground egress to the destination. Its aircraft mission profile separated takeoff, climb, cruise, descent, and landing, with taxi time at both ends and an added transition phase for tilt-rotor, tilt-wing, and tilt-duct aircraft. The same study assumed early airport-shuttle and air-taxi operations would use an onboard pilot and that the longest single-charge mission would be 50 miles.²⁹

NASA's later passenger-carrying OpsCon reinforces the same operational logic: early UAM depends on existing heliports and tightly controlled, low-volume operations, while intermediate intra-metro air shuttle service concentrates first on high-demand routes between public transportation centers, airports, and commuter stations. For scheduled shuttle missions, route structure supports air-traffic management through strategic trajectory planning, procedural separation, collision avoidance, time-space deconfliction, and flight-profile adjustment based on demand and environmental conditions. Thus, the reference missions function as system probes: each route tests whether access time, TOLA availability, aircraft range, passenger processing, charging, route structure, and airspace management can combine into a trip that is faster and operationally credible, not merely flyable.³⁰

Network impacts of mission ConOps

Mission ConOps shape the network before aircraft are dispatched: they determine which trips are eligible, which OD pairs are worth serving, how many vehicles are implied, and where infrastructure must absorb demand. Forecasting evidence supports this route-level framing because AAM demand estimates vary widely, no single passenger-demand forecasting method works best across scenarios, and bottom-up route methods require granular origin-destination data and more computation than top-down estimates. The practical implication is that network design should not begin with a generic citywide demand total; it should screen trips by travel time, trip length, OD density, service quality, and infrastructure reach.¹⁴

The strongest ConOps filter is whether flight can overcome the access, waiting, and transfer penalties around the air segment. Prior studies cited in the forecasting literature found that air taxi trips become competitive above different thresholds: longer than 10 km in one study, 15-25 km in another, and more than 30 minutes or more than 45 minutes for specific ground-trip substitution cases in another. A NASA-commissioned study cited in the same review found that 98% of air taxi demand would come from ground trips exceeding 30 minutes, while one constrained scenario estimated 82,000 daily passengers served by 4,100 four- or five-seat aircraft across ten U.S. metropolitan regions. These figures show why ConOps should be modeled as a network-trimming problem: eligibility thresholds can radically change fleet size, vertiport utilization, and corridor density.¹⁴

Aircraft design assumptions then feed back into the network. NDARC and SUAVE differed by about 7% in gross takeoff weight and showed substantial differences across major components, indicating high uncertainty in early eVTOL sizing. The same comparison found that battery and motor together represented around 40% of gross takeoff weight, while cruise consumed about 75-80% of mission energy in both tools. Because AAM missions have shorter range than conventional passenger aircraft, a smaller share of each trip is spent in efficient cruise, making takeoff, climb, and descent performance more important to vehicle sizing and network economics.⁷

ConOps also affect feasible infrastructure use. Orlando's Geico Garage analysis assumed a downtown vertiport as part of a 14-vertiport Florida network and used a \$30 million retrofit capital-cost assumption covering charging, passenger and crew facilities, maintenance repair and overhaul, vehicle gates, structural reinforcement contingency, and related facilities. The same analysis warned that the garage was not originally designed for rooftop vertiport structures and would require a detailed structural assessment to understand retrofit cost and risk. This illustrates a broader network issue: existing sites may reduce siting friction, but rooftop conversion can shift costs into structure, charging, and lost alternative use.⁴

Economic impacts scale with the operating network rather than the pad alone. For the Orlando vertiport, the first full year of operations was associated with approximately \$52 million in passenger revenue and \$35.88 million in total GDP impact across the City of Orlando, Orange County, and Florida. The same assessment separated direct, indirect, and induced effects, with total operational GDP impacts of \$22.32 million in Orlando, \$8.26 million in Orange County, and \$5.30 million in Florida. Oklahoma's statewide AAM analysis similarly estimated \$5 billion in GDP impact over 22 years, split into \$2.3 billion direct, \$1.3 billion indirect, and \$1.4 billion induced effects. These estimates are not transferable forecasts for every region, but they show that ConOps choices about passenger, cargo, vertiport, and manufacturing activity determine where economic value is counted.⁴³¹

The network consequence is that mission ConOps should be evaluated as coupled operational assumptions, not isolated flight paths. Demand thresholds determine which trips enter the network; aircraft performance determines range, payload, charging cadence, and reserve margins; infrastructure assumptions determine feasible nodes and retrofit costs; and economic models translate utilization into local, regional, and statewide impacts. A defensible AAM network therefore needs scenario-specific ConOps that explicitly connect trip selection, aircraft sizing, vertiport siting, fleet requirements, and economic leakage rather than treating demand, vehicles, and infrastructure as independent modules.¹⁴⁷⁴

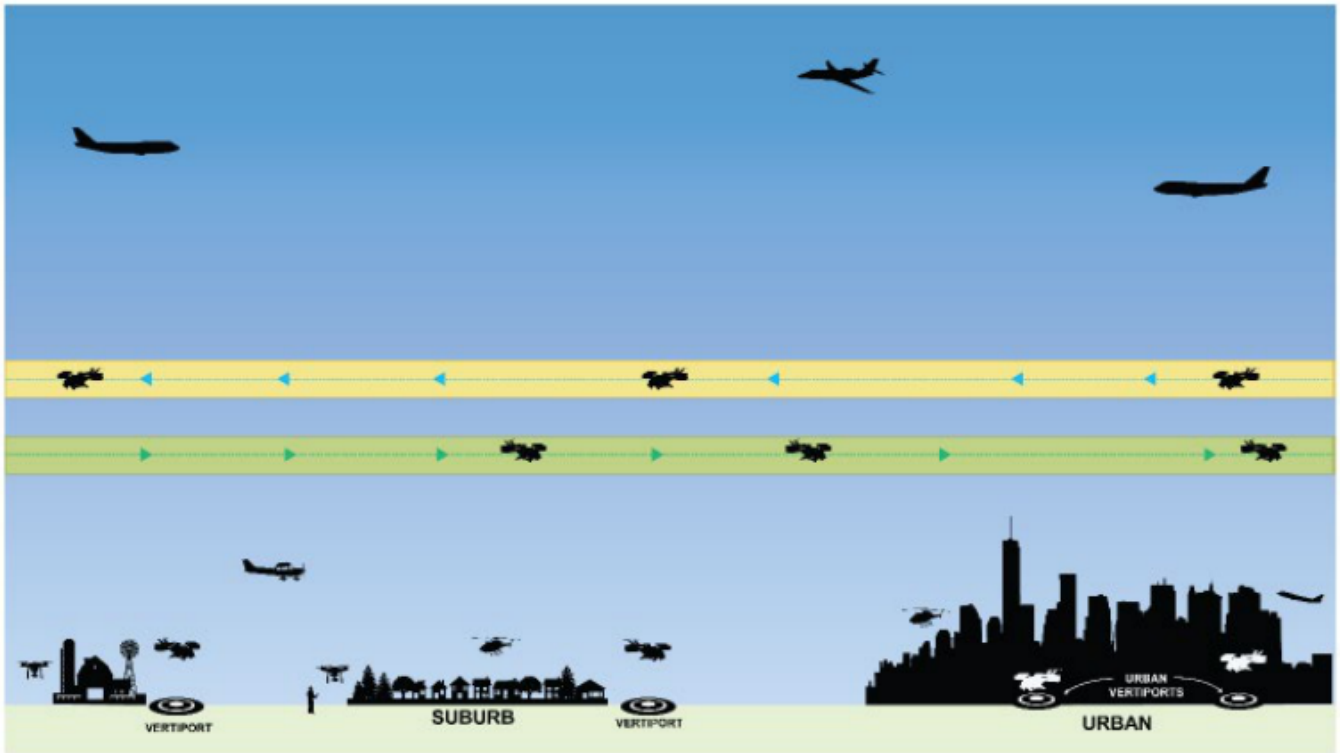


Exhibit. 57 Figure 33. Graph. Early stages of UAM operations. Source: S0012.

Vehicle Technology and Performance

Vehicle development, production, and technology gaps

Vehicle readiness is not just an airframe question. AAM is defined as aircraft with advanced technologies, including electric aircraft and eVTOLs, operating in controlled and uncontrolled NAS airspace; eVTOLs use electric power to hover, take off, and land, with configurations such as Lift + Cruise, Multiple Rotor, and Tilt Wing. That makes vehicle development inseparable from ConOps, certification standards, maintenance, infrastructure, and airspace integration: ConOps are meant to define operational use, requirements, architecture, nominal and contingency planning, and validation criteria.¹

The most material technology gap is automation. Oklahoma's 22-year AAM forecast assumes certified eVTOL aircraft by 2026, agreed ConOps and UTM standards by the end of 2024, service launches from 2026-2027, and a major automation-driven cost inflection around 2030. The same forecast states that automation is expected to reduce pilot costs, free a seat for revenue service, add safety benefits, and materially lower passenger tickets. That assumption is central, not incidental: the economic impact model explicitly factors automation into the AAM business case.³¹

Production location is another gap because local economic benefits depend on where vehicles are manufactured. Orlando's Geico Garage analysis excluded eVTOL vehicle CapEx because the aircraft were expected to be manufactured in Germany. Oklahoma's analysis, by contrast, estimated that an AAM OEM factory in the state would add \$2.2 billion in economic activity. The implication is straightforward: regions can host operations without capturing vehicle-production value.⁴³¹

Vehicle economics also depend on unresolved operating-cost assumptions. Orlando's model treated costs as half fixed and half variable, with variable elements including maintenance and spares, depreciation, customer acquisition, battery replacement, and energy. Oklahoma similarly separated AAM financial estimates into CAPEX, OPEX, revenues, and aircraft fleet purchases, with fleet purchases covering acquisition and maintenance costs needed to sustain identified use cases. Until battery replacement, maintenance, utilization, and fleet-purchase assumptions are validated in service, aircraft-level cost projections remain scenario-dependent.⁴³¹

Finally, vehicle development must close the standards and continuing-airworthiness path. A standard includes approved or acceptable manufacturing, design, maintenance, quality, production acceptance, operating, inspection,

identification, repair, alteration, and continued-airworthiness methods. Maintenance includes repair, adaptation, upgrade, modification, and preventive maintenance of NAS equipment or facilities. In practical terms, AAM vehicle development is not mature when prototypes fly; it is mature when aircraft, production systems, maintenance regimes, automation, and airspace interfaces can scale without breaking the operating case.¹

NDARC and SUAVE methodology comparison

The NDARC/SUAVE comparison shows that similar top-line aircraft sizing can hide large disciplinary disagreement. In the baseline case, NDARC estimated GTOW at 2,233 lb versus 2,086 lb in SUAVE, a difference of about 7%, while cruise L/D was 11.5 in NDARC versus 9.70 in SUAVE. The authors characterize NDARC's aerodynamic estimates as more optimistic, with NDARC's cruise L/D nearly 20% higher, while SUAVE used a more standardized aerodynamic computation.⁷

The disagreement is material because it appears across component weights and drag, not just in one parameter. SUAVE predicted higher total cruise drag than NDARC, including higher parasite and induced drag; its landing-gear drag estimate was more than twice NDARC's, and the total cruise drag difference between the tools was 22%. Weight estimates also diverged: NDARC estimated structural weight at 715 lb versus 505 lb in SUAVE, battery weight at 372 lb versus 444 lb, fuselage weight at 167 lb versus 127 lb, and landing gear at 141 lb versus 37 lb.⁷

The comparison also identifies which uncertainties matter most for eVTOL economics. Both tools found that the motor and battery together were about 40% of gross takeoff weight, and both predicted cruise to consume roughly 75-80% of mission energy. Yet the paper emphasizes that eVTOL range is much lower than conventional passenger-aircraft range, making takeoff, climb, and descent performance comparatively more important than in longer-range aircraft. Hover and climb modeling also affects sizing of wiring, motors, and in some cases batteries, so low-speed and vertical-flight assumptions are not secondary details.⁷

Trade studies reinforce the same point: NDARC and SUAVE can disagree on design direction, not merely magnitude. In rotor-radius sweeps, both tools predicted lighter aircraft as rotor radius shrank, although NDARC showed a local optimum; SUAVE estimated hover tip speed at 470 ft/s versus 650 ft/s in NDARC, creating a discrepancy the authors said should be followed up in an actual aircraft design program. In wingspan sweeps, the tools predicted opposite GTOW trends because SUAVE treated wing structural loading and rotor loads along the span differently than NDARC.⁷

For system modeling, the implication is that aircraft outputs should be treated as scenario-sensitive inputs, not fixed constants. Fleet-optimization frameworks represent trips, source/sink links, and feasible task-to-task transitions, with transition feasibility depending on whether a prior task end time plus transition time is no later than the next task start time. If aircraft weight, drag, power, energy use, or transition timing shifts across design tools, the feasible network, fleet requirement, utilization, and operating cost estimates can shift with it.²⁵⁷

Powertrain modeling, topology selection, and validation

Powertrain assumptions should be embedded in aircraft sizing, not treated as a fixed efficiency multiplier. Earlier electric NDARC modifications modeled batteries and motors with peak power requirements, conversion efficiencies, and mission energy, using constant motor efficiency of 95%, 3% losses for power conversion and conditioning hardware, and 2% battery conversion losses. That approach also required replacing the fuel-burn mission iteration with an energy-balance formulation, converging gross weight, empty weight, and battery capacity for electric aircraft sizing.³²

More detailed powertrain optimization adds topology, thermal, passive-component, and mission-profile decisions to the sizing loop. Cheng et al. evaluated three inverter topologies for eVTOL powertrains: two-level, three-level active neutral point clamping, and three-level T-type inverters, all using sinusoidal pulse width modulation. The study limited semiconductor selection to wide-bandgap devices because of their efficiency, voltage rating, and high-frequency operation, while optimizing converter technology, voltage rating, and the number of parallel devices.¹⁵

The component model matters because mass and losses move across the aircraft system. Cheng et al. considered ceramic and film capacitors for high power density and low weight; modeled inductor core and winding design; compared magnetic materials, core shapes, and operating conditions; used magnetic equivalent circuits and nonlinear iteration to estimate flux density; and applied the improved generalized Steinmetz equation for core losses. The winding model used Litz wire because of lower winding losses, and the cooling model compared forced-air and

single-phase liquid cooling to minimize heatsink mass while meeting thermal constraints. The optimizer used a hybrid genetic algorithm and simulated annealing method to minimize eVTOL cost per mission across powertrain and mission-profile degrees of freedom for a NASA lift+cruise configuration.¹⁵

The Los Angeles validation-style case illustrates why route definition and powertrain design cannot be separated. Cheng et al. selected an LAX-to-Burbank revenue mission, using a 30 km flight distance, 15 minutes of flight time, and 15 minutes for boarding, for a roughly 30-minute single-trip mission. The optimization converged to \$183.5 per mission, selected the 3L-ANPC inverter, and converged at 111 kHz switching frequency, 1400 V DC bus voltage, 30 km maximum mission range, and 589 kg battery mass, with no final constraint violations. The same result identified hover as the highest-power and lowest-efficiency segment, while reserve had the lowest required power.¹⁵

Sensitivity analysis reinforces the planning implication: vehicle economics are route-sensitive. As cruise range increased, the powertrain's effect on direct operating cost and gross aircraft mass became more significant, while battery mass rose with the added energy requirement. Because voltage had already reached its preset maximum, longer-range cases increased current, requiring thicker wires and stronger cooling to satisfy heat-dissipation and current-density constraints. The optimizer then reduced switching frequency to offset increased conduction losses, showing that topology settings respond to mission range rather than remaining fixed.¹⁵

Battery improvements help, but not without limits. Cheng et al. found that higher battery energy density reduced direct operating cost, gross mass, output power, battery mass, and powertrain mass, but with diminishing reductions as battery energy density increased. Battery life cycle affected battery cost but did not change the optimal powertrain design. For feasibility screening, this means powertrain validation should report not only nominal range and efficiency, but the coupled values that drive network models: mission-specific power by segment, battery mass, converter topology, voltage, switching frequency, cooling approach, and direct operating cost.¹⁵

Mass, mission, and direct operating cost models

Mass and cost results for eVTOLs should be read as mission-specific outputs, not generic vehicle properties. The available literature notes that evidence-based VTOL performance studies remain limited because regular urban public or personal VTOL use is still new, and many publications focus on potential applications and requirements rather than measured operations. This makes assumptions about mission length, charging, utilization, battery life, and maintenance central to any direct operating cost estimate.²²

Cheng et al.'s cost-optimization model ties direct operating cost to powertrain design and the mission profile. The model includes battery replacement, airframe depreciation, maintenance, pilot cost, and electricity cost; it assumes 2,000 operating hours per year, a five-year aircraft life, 10,000 total service hours, 10% residual value for airframe and avionics, \$200/kWh battery cost, 40% battery residual value, and 1,000 battery cycles to 70%-80% state of health. Maintenance is modeled through maintenance labor hours per flight hour and mechanic wages, while flight cost includes pilot wages and annual electricity use estimated through the powertrain model.¹⁵

The same study applied the model to a Los Angeles airport-transfer mission between LAX and Burbank, using a 30 km revenue mission, 15 minutes of flight time, 15 minutes of boarding time, and roughly 30 minutes total mission time. The optimization converged to \$183.5 per mission, selected a 3L-ANPC inverter, set switching frequency at 111 kHz and DC bus voltage at 1,400 V, matched maximum mission range to the 30 km revenue mission, and produced a 589 kg battery mass. Within that mission, hover had the highest required power and lowest efficiency, while the reserve segment had the lowest required power.¹⁵

Sensitivity results show why mission range cannot be separated from aircraft mass and DOC. As cruise mission range increases, DOC, gross mass, battery mass, and the influence of the powertrain model all become more significant; cruise range particularly affects electricity cost and battery depreciation cost. At longer ranges, higher battery and aircraft body weight require higher motor output; once voltage reaches its preset maximum, current rises, thicker wiring and stronger cooling are needed, conduction losses increase, and the optimizer lowers switching frequency to preserve powertrain efficiency. Higher battery energy density reduces DOC, gross mass, depreciation cost, output power, battery mass, and powertrain mass, but with diminishing returns as airframe, maintenance labor, and pilot salaries become more dominant cost contributors.¹⁵

Other LCA-oriented assumptions illustrate how sensitive these models are to input choices. One eVTOL LCA assumed \$700/kWh batteries, 230 Wh/kg battery output, 2,000 battery cycles, one hour to charge a 100 kWh battery, \$70/kg propulsion motors, 5 kW/kg motor capability, 6,000-hour motor and servo lifetimes, 0.1 maintenance person-

hours per flight-hour, 90% charging efficiency, and 12 cents/kWh average U.S. electricity cost. That same study treated environmental externalities as including battery and vehicle-parts disposal, noise, power-based emissions, and manufacturing and maintenance emissions. The implication for feasibility modeling is direct: mass, mission profile, battery replacement, charging time, powertrain topology, utilization, and residual value must be modeled together before DOC estimates can be used for network planning.²²

Electric aircraft performance and operating economics

Electric aircraft economics should be evaluated at the mission and network level, not as a vehicle-only metric. Route geometry can change both performance and acceptance: in the San Diego-Los Angeles reference mission, the direct route was shortest, but involved open-water exposure and low-altitude overflight of Los Angeles communities below 2,000 ft AGL; the water route was 15% longer by distance and 11% longer by duration, while operating 71% of its distance over open water and using a higher-altitude VFR corridor through LAX Class B airspace. The inland route increased distance by 39% and duration by 32%, but reduced exposure to dense population, noise, congestion, and coastal hazards.⁶

This means “performance” includes climb, cruise, routing, altitude, ATC access, certification exposure, and community impact. Noise was identified as perhaps the most significant near-term constraint for ODM Aviation, with electric motors, distributed propulsion, frequency spreading, and tiltrotor configurations offering potential reductions relative to helicopters. Aircraft capabilities also affect operational mitigation: higher cruise altitudes, extended vertical ascents and descents, avoidance of impulsive noise, and routing around noise-sensitive communities may reduce ground impacts.⁶

Direct operating cost depends heavily on assumptions that remain uncertain. One eVTOL DOC model used airframe purchase price of \$1,102/kg, avionics purchase price of \$100,000, battery purchase price of \$200/kWh, electricity at \$0.16, 1,000 battery cycles, 40% battery residual value, one pilot per aircraft, pilot wrap rate of \$134/hour, mechanic wrap rate of \$62/hour, and maintenance at 60% per flight hour. Broader market analysis similarly frames viability around acquisition price, maintenance expense, operating cost, passenger occupancy, and avoiding empty repositioning trips.¹⁵³³

The minimum viable aircraft is therefore not just “electric” or “VTOL.” Bain estimates that operators need aircraft with roughly 70-80 miles of range or more and capacity for at least three to four passengers, while four- or five-seat eVTOLs are expected to cost about \$3 million to \$6 million and require at least 1,000 flight hours per year for economic viability. Achieving that utilization also depends on fast charging, IFR capability, night and poor-weather operations, and all-weather aircraft performance.³³

The practical conclusion is that electric aircraft operating economics are constrained by coupled variables: battery cost and life, payload, range, hover time, route restrictions, charging time, pilot cost, maintenance, vertiport access, and airspace integration. Simulation is likely necessary because full-scale air taxis and high-density vertidromes are not yet available; DLR notes that vertidrome and network research currently relies on simulation, with recharge time materially affecting fleet and ground-infrastructure requirements.¹⁸

Telecommunications and autonomy for ODM aviation

Telecommunications and autonomy should be treated as part of the operating architecture, not as accessory vehicle features. Vascik’s ODM ConOps explicitly includes customer ordering, aircraft staging and ferrying, boarding, flight-path planning, ATC interaction, and charging, meaning the communications layer spans both passenger service and aircraft operations.⁶

The immediate operational requirement is reliable interaction with controlled airspace. In the San Diego-Los Angeles reference mission, all flights require entry into San Diego Class B airspace after takeoff and must request routing from ATC to operate in and exit that airspace. Route choice changes the communications burden: the Malibu-Century City water route requires ATC access to Santa Monica Class D during operating hours, and the “pilot or automation” must establish and maintain two-way radio contact with Santa Monica tower between 1500-0500Z. Similar requirements recur in longer profiles, where descending below specified altitudes can trigger two-way radio communication with control towers and where helicopter VFR routes create congestion considerations.⁶

At scale, the problem shifts from individual radio contact to networked traffic management. NAS-dependent UAM constraints include weather restrictions, access to controlled airspace, autonomy interactions with ATC, and safety

in congested areas, with principal impacts on ATC scalability, all-weather operations, and network operations. FAA-oriented concepts assume UAM aircraft operate within the NAS but separated from other users through corridor networks, with data shared through FAA-NAS exchange protocols to connect the FAA and UAM community. Proposed static and dynamic airspace cutouts could allocate airspace procedurally, with ATC defining availability while operations inside the cutout are not directly controlled by ATC.¹⁰

Autonomy therefore has to be evaluated against specific functions. The UAM literature identifies detect-and-avoid, automated mission management, vehicle health management, simplified vehicle operations, and electric power as priority aircraft sub-systems. Detect-and-avoid remains a gap because a performance-based standard for UAM DAA has not yet been put in place, and existing onboard systems are less powerful than ground-based radar. Automated mission management would control mission-critical functions and requires stronger real-time data processing to achieve algorithm reliability. Vehicle health management also becomes operationally central because UAM fleets are expected to feed real-time health assessments into AMM and UTM systems, rather than using VHM only as a supplemental aircraft function.³⁴

The autonomy roadmap should be staged. Mature UAM forecasts assume movement from full cockpit, to single pilot, to fully autonomous vehicle operations, but that sequence depends on both autonomous aviation technology and electric propulsion readiness. Certification is a binding constraint: current certification methods test components against deterministic requirements, which makes highly integrated or nondeterministic systems harder to certify. Electric or hybrid propulsion and advanced fly-by-wire systems with greater control authority add similar certification issues, so early designs will likely rely on tailored existing standards and special requirements rather than a mature, standardized autonomy pathway.^{34 30}

For feasibility analysis, the practical implication is direct: ODM autonomy claims should be converted into testable operational assumptions. A defensible model should specify who communicates with ATC, when two-way contact is required, what data are exchanged with NAS systems, how DAA is validated, how AMM handles rerouting and airspace constraints, and how VHM feeds dispatch, maintenance, and traffic-management decisions. Without those assumptions, autonomy is an aspiration rather than an operating concept.



Exhibit. Figure 14 portrays mission profiles for the Intra-Metro Air Shuttle mission. Source: S0039.

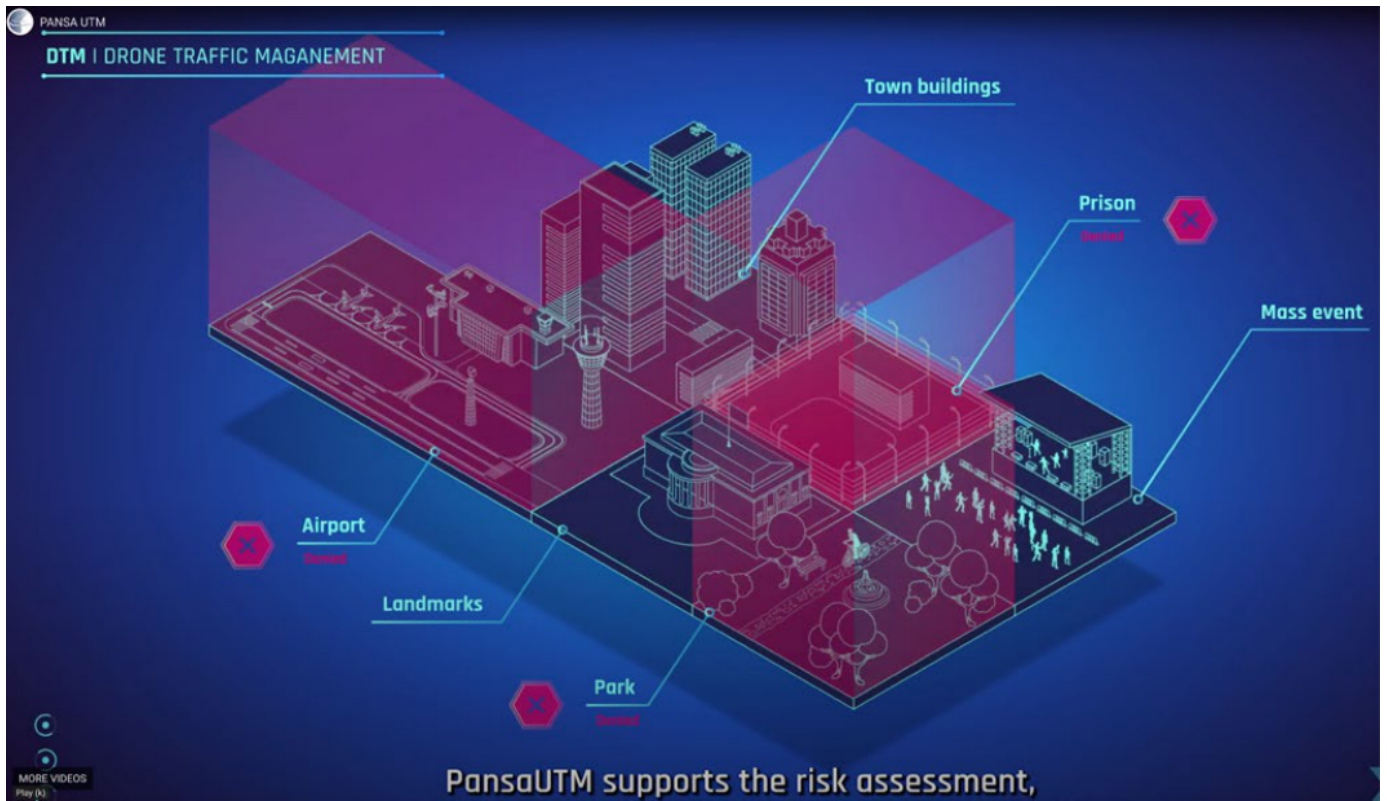


Exhibit. Figure 32: Example of managing the urban airspace operations (e. Source: S0076.

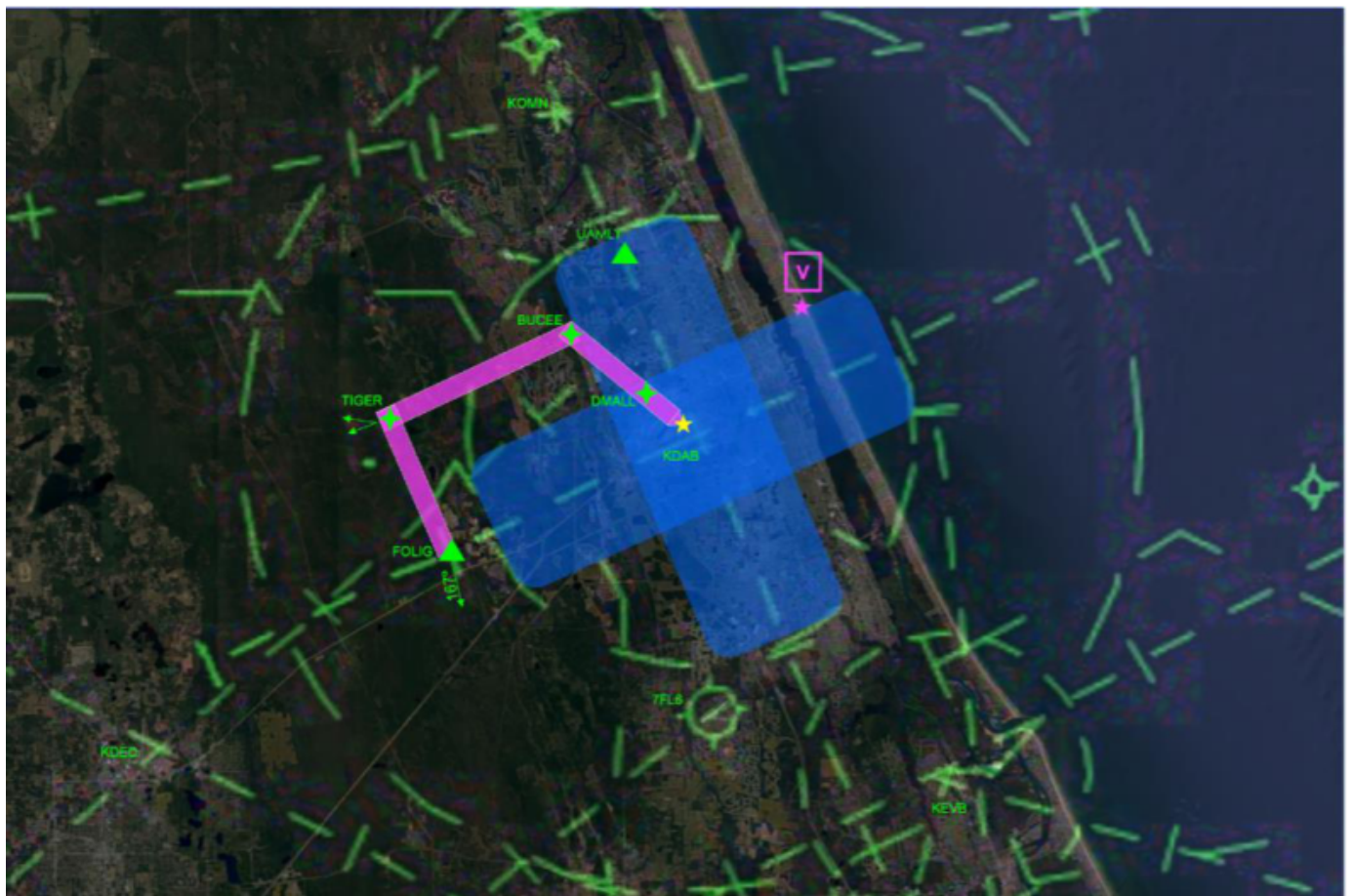


Exhibit. Figure 4.8. A proposed concept of the FOLIG Transition. Source: S0061.

Infrastructure and Electrification

Existing facilities and planned upgrades

Existing aviation infrastructure is a starting point for AAM, not a ready network. Vascik's Los Angeles work treats takeoff-and-landing-area availability as a core operational constraint and separates existing aviation ground infrastructure from the additional work needed to expand TOLA availability, including aircraft performance limits, regulation, community acceptance, and business feasibility. The same source identifies Los Angeles figures on the geographic location and type of existing aviation ground infrastructure and notes that existing heliport infrastructure drops off outside the central business district, implying that legacy facilities are spatially uneven for a metropolitan ODM network.⁶

Airport electrification is more concrete where facilities already have charging plans. In the PennDOT AAM and Airport Electrification study, PHL's listed airside charging assets include one Level 2 port at the Fleet Shop, four Level 2 ports at the Maintenance Service Building, two planned DCFC ports at the Maintenance Service Building, and three planned DCFC ports on the airfield. The same report separately identifies a table for "PHL Current and Planned Landside EV Charging," indicating that airport electrification planning spans both airside and landside use cases.²⁰

Retrofitting non-aviation structures can create capacity, but it shifts risk into structural, commercial, and operating assumptions. Orlando's conceptual Geico Garage vertiport used a \$30 million retrofit capital-cost assumption covering electric charging infrastructure, passenger and crew facilities, maintenance repair and overhaul facilities, vehicle gates, and contingency for structural reinforcement. Because the garage was not originally designed for rooftop vertiport structures, the study stated that structural reinforcement would likely be necessary and that a detailed structural assessment would be needed to understand cost implications and risks. The same case estimated \$600,000 to \$1 million in annual lost parking revenue from converting the roof deck, while suggesting a potential \$2.1 million annual ground lease to an AAM operator, though that lease was not included in the IMPLAN inputs.⁴

Planning also has to treat vertiports as operational nodes, not isolated pads. DLR's HorizonUAM work describes vertidromes as hubs where users enter and exit the UAM network, and states that their design affects fleet composition, air traffic management, efficiency, and throughput. Because full-scale air taxis and high-density vertidromes are not yet available, DLR emphasizes simulation, validated by flight trials, for vertidrome and network planning. Its reported tools include micro-level vertidrome simulation, agent-based vertidrome-network management and optimization, and human-in-the-loop simulation for controlled-airspace and airport-environment workload.¹⁸

The implication for near-term AAM is straightforward: prioritize facilities where aviation access, electrical capacity, passenger processing, maintenance, and airspace procedures can be upgraded together. Existing airports, heliports, garages, and modular test facilities may each contribute, but none removes the need for integrated planning across charging, structural suitability, throughput, ATC workload, and community-facing constraints.²⁶¹⁸

Vertiport and TOLA availability, access, and siting

Vertiport siting is less a real-estate exercise than a network-integration problem. Local planning must account for approach and departure airspace, adjacent structure heights, noise-sensitive land uses, historically marginalized communities, first- and last-mile access, transit hubs, and energy availability. Because most VTOL concepts are expected to be electric, hydrogen-powered, or hybrid-electric, siting also depends on local electrical capacity, hydrogen availability, and the ability to absorb AAM demand alongside broader EV growth.³⁵

Existing aviation infrastructure can reduce deployment friction, but it does not eliminate siting constraints. Current FAA treatment, as summarized in *Beyond Takeoff*, reduces the distinction between heliports and vertiports by treating them as part of unified vertical-lift infrastructure; the same source states that FAA Engineering Brief 105A classifies vertiports as a type of heliport. This creates an opportunity to adapt existing heliport assets through common standards for weight limits, aerodynamic parameters, and operating protocols, potentially reducing new construction, environmental impact, and cost. The constraint is that legacy facilities may be poorly matched to future demand geography, multimodal access, community tolerance, or charging requirements.⁵

The strongest siting principle across the evidence is co-location with existing transportation nodes. Miami-Dade's UAM roadmap recommends locating vertiports with transit hubs and first-/last-mile transportation, identifying airport, rail, people-mover, and park-and-ride connections as examples of multimodal opportunities. This is not only a planning preference; demand modeling evidence indicates that vertiport prevalence and accessibility affect realized

demand, and that population distribution and vertiport catchment areas can materially change market outcomes. In practical terms, a technically feasible landing site with poor access may still fail as a mobility product. [3517](#)

Land-use authority is therefore central. Local governments shape vertiport zoning and land-use policy, while federal and state agencies provide coordination across developers, planners, and regional stakeholders. Dedicated vertiport zoning can address operating needs that do not fit traditional zoning categories, and overlay zones can encourage development near transit hubs while preserving airspace access through aviation easements and height restrictions. These same policies should protect sensitive sites such as schools and hospitals through conditional permits, noise mitigation, and environmental safeguards. [5](#)

Community impact is a siting constraint, not a post-approval communication task. Miami-Dade identifies noise-sensitive residential and institutional uses, visual pollution, safety and security, social equity, and potential disproportionate impacts as planning considerations. A broader AAM evidence review similarly lists safety, noise reduction, emissions, privacy, land use, and visual pollution as key social acceptance issues, with infrastructure limitations among the primary obstacles for potential users. This means vertiport availability should be measured not just by physical pads or rooftops, but by publicly acceptable, legally permitted, accessible, energy-ready sites. [3526](#)

Retrofitting existing structures can be expensive and site-specific. Orlando's conceptual downtown vertiport assumed a \$30 million retrofit of the Geico Garage, including charging infrastructure, passenger and crew facilities, maintenance facilities, vehicle gates, contingency, and likely structural reinforcement because the roof deck was not originally designed for vertiport use. The same analysis estimated \$600,000 to \$1 million in annual lost parking revenue from repurposing 300 roof-deck spaces for about 100 events per year, while suggesting a possible \$2.1 million annual ground lease from an AAM operator could offset that loss. These assumptions show why rooftop availability cannot be inferred from map geometry alone. [4](#)

For public agencies, the near-term path is targeted pilots rather than broad network buildout. PennDOT's study used demographic and economic indicators, transportation assets, and health or emergency access needs to build hot-spot maps, then refined those maps through stakeholder engagement and interagency collaboration. Its highest-ranked strategic investment priorities were rapid incident response, rural emergency services, remote medical delivery, airport shuttle, urban-core-to-urban-core transport, and corridor freight support. That prioritization implies that early vertiport and TOLA investments should be tied to use cases with public benefit, existing infrastructure leverage, or clear emergency/logistics value. [20](#)

Funding reinforces this targeted approach. PennDOT's funding review found 85 programs across federal, state, regional, and private sources, with AAM funding able to support on- and off-airport infrastructure such as urban vertiports, multimodal hubs, and last-mile connectivity. Airport electrification funding is narrower, focused on on-airport clean energy, GSE electrification, zero-emission vehicles, and power-system upgrades. For vertiport siting, the implication is straightforward: airport-based TOLAs may compete through airport and electrification programs, while urban or multimodal vertiports will likely need broader AAM, transportation, economic-development, and local funding packages. [20](#)



Exhibit. Figure 19: Uber Elevate Vertiport Infrastructure Rendering Source: Uber Elevate Whitepaper, <https://d1nyezh1ys8wfo>. Source: S0050.

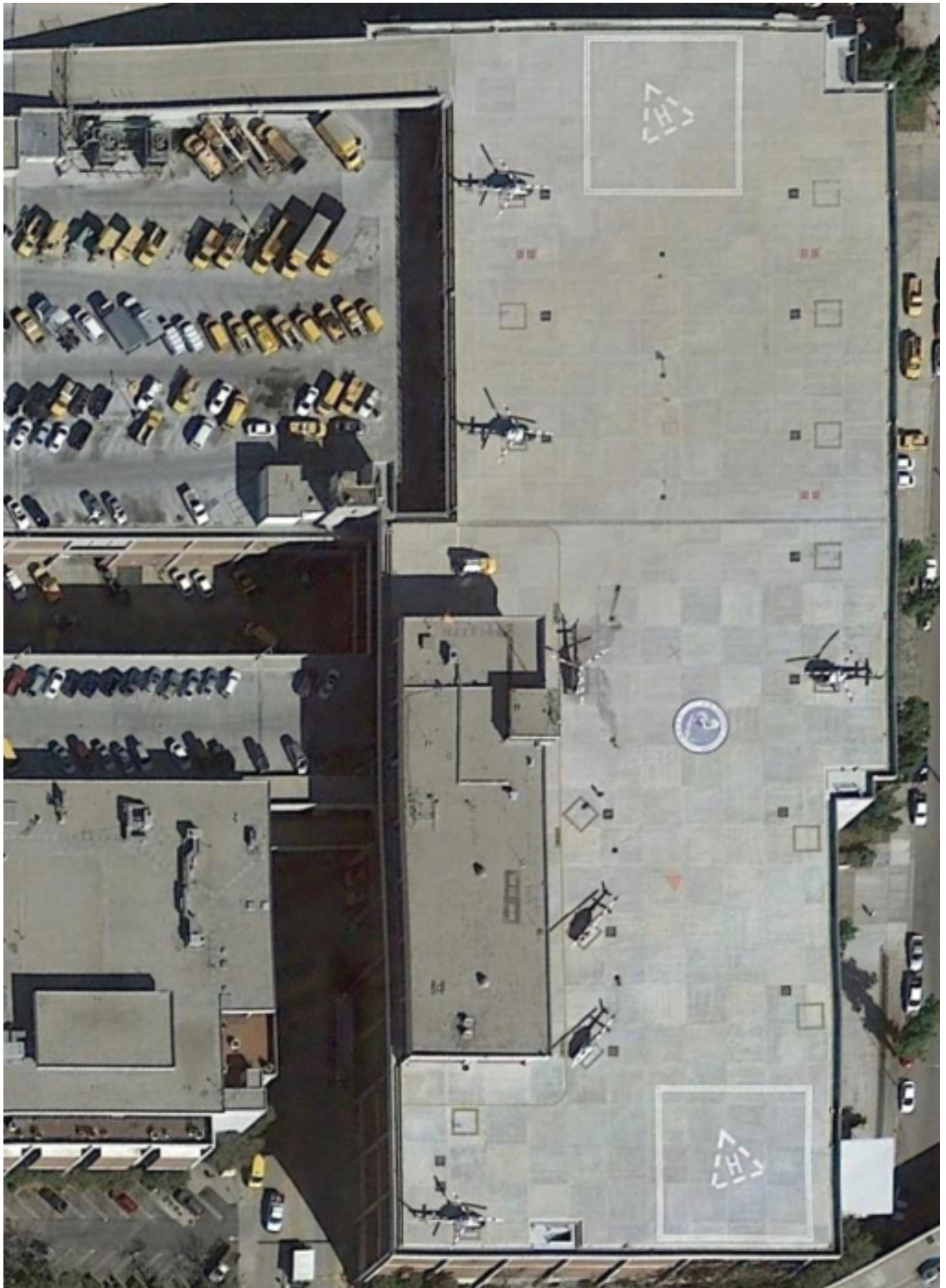


Exhibit. Single-Pad TOLA High-Capacity TOLA. Source: S0011.

Airport electrification initiatives

Airport electrification is emerging as the practical bridge between today's airport infrastructure and future AAM/RAM operations. PennDOT's feasibility study treated airport electrification as a complementary enabling component for AAM deployment, alongside use-case evaluation, infrastructure needs, policy issues, funding opportunities, and stakeholder-informed project prioritization. Its assessment included electrification initiatives at three Pennsylvania airports and fed into a broader roadmap for PennDOT planning, investment, and coordination.²⁰

The most mature near-term initiatives focus on conventional airport energy uses: electric ground support equipment, gate power, pre-conditioned air, recharging stations, EV integration, clean energy systems, and airport power infrastructure. This matters because these investments can decarbonize existing airport activity before AAM demand is proven, while also preparing electrical capacity, operational procedures, and grant-administration experience that future electric aircraft may require. PennDOT's funding review found 46 funding opportunities applicable to airport electrification after narrowing from broader AAM-aligned programs and adding clean energy, environmental justice, and sustainable transportation sources.²⁰

Federal funding is already central to this transition. PennDOT identified the FAA Airport Improvement Program, VALE, Airport Terminal Program, USDOT SMART, RAISE, EDA tools, and newer FAA Reauthorization Act authorities as relevant pathways for airport electrification or adjacent clean transportation investments. AIP remains the backbone: it allocates about \$3.2 billion annually for airport capital projects, distributed more than \$10 billion nationally from 2022 to 2024, and sent more than \$268 million to Pennsylvania airports during that period. During the same period, AIP funded more than \$73 million in electrification-related projects, including GSE electrification, gate systems, and recharging stations. VALE has also supported Pennsylvania investments above \$32 million from 2008 to 2023 for pre-conditioned air, ground power, recharging infrastructure, and electric vehicles at airports including PHL and PIT.²⁰

Washington's approach shows the network dimension more explicitly. WSDOT's 2020 Electric Aircraft Feasibility Study screened 134 public-use airports for beta test sites for electric aircraft infrastructure, considering runway length, aviation service need, airport connectivity within 500 nautical miles, on-airport aerospace manufacturing, fixed-base operators, geographic dispersion, and jet fuel availability for hybrid-electric aircraft. WSDOT later concluded that a single electrified airport has limited utility beyond pilot training; useful regional air mobility depends on a network of airports that aligns with passenger and cargo needs, energy infrastructure, transportation corridors, and community economic goals.⁸

That network logic shifts airport electrification from a facility upgrade to a corridor strategy. WSDOT argues that RAM can complement trucking, rail, road, and sea by connecting nodes in a broader freight and mobility system, and that co-locating modes at airports can concentrate energy infrastructure, vehicle infrastructure, and maintenance activity. Washington initiatives include an almost \$1 million federal grant for a Chehalis-Centralia Airport hydrogen fueling feasibility study and a six-airport federal funding effort for electric aircraft charging infrastructure. WSDOT frames these multi-airport energy efforts as potentially beneficial to both individual airports and the regional transportation network.⁸

The main implementation lesson is sequencing. Airport electrification should begin with defensible on-airport uses that can stand on their own, then scale toward electric aircraft charging only where route demand, grid capacity, multimodal access, and funding eligibility converge. Existing airport programs can fund near-term decarbonization, while state and regional planning can identify the airport networks that make electric aviation operationally useful rather than symbolic.²⁰⁸

Electrification requirements and future conversion to electric

Electrification is not just charger installation; it is a conversion of aviation sites into energy nodes. AAM aircraft are expected to use batteries or hydrogen fuel cells to enable reduced- or zero-emission aviation, and WSDOT frames that shift as part of a broader transportation mode connecting aviation with other modes and geographies. Near-term implementation can reuse heliports, commercial service airports, and underutilized general aviation airports, but WSDOT expects modifications or enhancements, with new infrastructure needed as aircraft types and off-airport use cases emerge.⁸

The core requirement is network readiness. WSDOT's 2020 electric aircraft feasibility work screened 134 public-use airports for beta electric-aircraft sites, using criteria such as a 3,000-foot runway, aviation service need, connectiv-

ity to airports within 500 nautical miles, on-airport aerospace manufacturing, fixed-base operator presence, geographic dispersion, and jet fuel availability for hybrid-electric aircraft. The later AAM plan qualifies those criteria: runway dependence may decline as STOL and VTOL concepts mature, but the conclusion that electric aviation needs a network, not isolated sites, remains central.⁸

A single electrified airport has limited value beyond training if surrounding airports cannot support passenger and cargo movement. Future conversion should therefore prioritize corridors and clusters where airports match passenger demand, cargo flows, energy infrastructure, and statewide transportation needs. In that model, regional air mobility can help catalyze clean energy corridors and broaden rural economic opportunity, while airports can become mobility, logistics, energy, and first-responder hubs rather than lightly used aviation-only assets.⁸

Electrification planning should also be multimodal. WSDOT identifies opportunities to co-locate air, truck, rail, road, sea, and last-mile delivery functions so that energy infrastructure and maintenance activity can be concentrated across modes. The Pacific Northwest examples point in this direction: Chehalis received an approximately \$1 million federal grant to study a multimodal hydrogen fueling facility, and a six-airport group led by Centralia-Chehalis Airport sought federal funding for electric aircraft charging infrastructure. WSDOT also notes a separate request to plan and construct 10 multimodal chargers at six public-use airports in and around the Seattle-Tacoma-Bellevue and Puget Sound region.⁸

The cost and design burden can be material. Orlando's conceptual Geico Garage vertiport assumed a \$30 million retrofit, including electric charging infrastructure, passenger and crew facilities, maintenance repair and overhaul facilities, vehicle gates, structural reinforcement contingency, and other vertiport-related improvements. Because the garage roof was not originally designed for vertiport structures, the study noted that structural reinforcement would likely be required and that a detailed structural assessment would be needed to understand cost and risk. This illustrates a broader conversion lesson: rooftop or urban sites may appear attractive geographically but still face expensive structural, operational, and revenue tradeoffs.⁴

Future conversion to electric should therefore proceed in stages: first, upgrade existing aviation sites that already have aviation governance and access; second, build airport clusters where charging or hydrogen capacity supports actual travel and freight corridors; third, extend infrastructure to new vertiports only where demand, access, structural feasibility, grid capacity, and community acceptance align. The practical test is whether a site strengthens the network, not whether it can host a charger.

Charging coordination, UAM charging demand, and power grid flow

UAM charging demand should be modeled as an operations problem, not as a static charger count. Demand depends on where trips concentrate, when aircraft arrive, state of charge, station capacity, and grid constraints. In the Dallas-Fort Worth airport-access case, a 50-vertiport network at \$2 per passenger-mile produced about 3,202 one-way UAM passenger trips, but demand was highly price- and access-sensitive: a \$0.50 CPM increase reduced demand by 26%, while adding vertiports from 50 to 75 raised demand only 12.5% and adding 75 to 100 raised it only 2%. This matters for electrification planning because incremental vertiports do not necessarily distribute load evenly; airport demand remained concentrated in corridors serving DFW, Dallas downtown, DAL, Fort Worth downtown, and Richardson.³⁶

Charging coordination links passenger demand to fleet dispatch. A Miami-Dade smart-charging study selected 70,000 likely UAM trips using saved travel time above 30 minutes and distance above 30 miles, divided the county into 27 vertiport stations, assumed a 1,500-aircraft fleet, and treated county charging demand as load on a grid bus. Its dispatch formulation tracks aircraft by vertiport, destination, departure and arrival time, and state-of-charge transitions, while also representing parked aircraft and charging from one SoC level to another. The same formulation constrains maximum charging capacity at each station, maximum charging power, over- and under-charged session demand, generation limits, transmission-line power-flow limits, and generator ramping limits.³⁷

The practical implication is that vertiports become grid-interactive assets. Charging can be shifted within operating windows, but only while preserving fleet readiness and passenger service. In the Miami-Dade setup, a 20-minute flexible time window was used for dispatch and charging coordination, and uncertainty was modeled around charging-session arrival time and energy demand. For airports, the same logic suggests that charger sizing should be tied to corridor peaks, terminal-access penalties, turnaround assumptions, and feasible dispatch slack, not average daily demand alone. Orlando's Geico Garage concept also treated electric charging infrastructure as part of a broad-

er vertiport retrofit package that included passenger and crew facilities, MRO facilities, vehicle gates, structural reinforcement contingency, and a \$30 million capital cost assumption.^{[374](#)}

Power system welfare and smart charging

Smart charging reframes UAM electrification as a joint transportation-grid optimization problem. A UAM operator must decide which day-ahead trip requests to accept, how to dispatch aircraft among fixed vertiports, and when to charge aircraft whose state of charge changes across parked, charging, and in-flight states. In this framing, the fleet is not merely new demand on the grid; it is a controllable load that can be shifted in time and space if charging is coordinated with dispatch.^{[37](#)}

The relevant objective is therefore not just minimizing aircraft charging cost. Wu et al. model both the UAM system and the power system, with charging sessions bounded by earliest/latest feasible charging windows, station capacity limits, generator limits, ramp constraints, external loads, voltage angles, and transmission flow limits. Their formulation explicitly seeks to maximize power system welfare while satisfying vertiport charging needs and grid operating constraints. This is the right level of abstraction for AAM planning because charging demand is created by the flight schedule, while grid impact depends on where and when those charging sessions occur.^{[37](#)}

Uncertainty makes this harder than a deterministic schedule suggests. Aircraft charging behavior is stochastic because weather, temperature, and emergencies can affect operations; deterministic models can therefore produce ineffective schedules. A chance-constrained approach addresses this by bounding the probability that a charging plan fails below a chosen confidence level, while preserving dispatch decisions that maximize fulfilled UAM demand. For early deployment, this implies that charging policy should be treated as an operating-control problem, not a static equipment-sizing exercise.^{[37](#)}

Airport examples show why this matters before large-scale AAM arrives. Harrisburg International Airport had one EV charging station that delivered 60 charging hours and 3,100 kWh in July 2024, planned a 2 MW solar farm for load shedding, and would need an additional 11,609 kWh per month to electrify listed gasoline/diesel equipment, a 12.7% increase over average monthly consumption. Johnstown-Cambria County Airport would need 25,742 additional kWh per month for comparable equipment conversion, equal to a 55.5% increase in average monthly consumption. These are ground-electrification cases, but they illustrate the same planning issue: electrification changes load shape, site capacity needs, and operating economics.^{[20](#)}

The infrastructure risk is that charging capability becomes the binding constraint on aircraft utilization. Evidence reviews identify grid access and charging infrastructure as technological barriers, noting that AAM operations depend heavily on battery charging and fast-charging capability, and that insufficient grid capacity can create grid stress, long waits, and inefficient operations. Industry interviewees in the same review emphasized that large electricity supply must be guaranteed and that weak electrical infrastructure erodes the AAM value proposition. Smart charging can reduce that risk, but only if vertiport siting, dispatch, charging windows, and grid constraints are planned as one coupled system.^{[17](#)}

Operations, Airspace, and Regulation

Aircraft, fleet, and traffic operations management

Aircraft operations management must bridge three domains: the individual aircraft, the operator's fleet, and the shared airspace. NASA's passenger-carrying UAM OpsCon separates "Individual Vehicle Management & Operations" from "Air Traffic & Fleet Operations Management," with FAA roles in airworthiness, operational approval, and safe ATM; industry roles in ground operations, maintenance, and fleet management; and NASA roles in efficient and scalable ATM research. This division matters because AAM maturity depends not only on certifying aircraft, but on coordinating dispatch, maintenance, weather, charging, port use, and separation at network scale.^{[30](#)}

Near-term operations are likely to use existing airports, heliports, procedures, and controlled-airspace structures where possible. WSDOT expects early AAM entry into service to rely on existing infrastructure, although airports, heliports, and parking structures are limited by location, size, structural capability, and AAM-specific needs such as charging, passenger processing, and high-speed data connections. At airports, VTOLs may use existing arrival and departure patterns, taxi or hover to designated parking/loading areas, and repurpose paved apron space for charging and passenger handling. Towered airports can integrate AAM traffic with conventional operations through air

traffic controllers, but vertiport ingress and egress routes must avoid runway operational areas and preserve aircraft separation.⁸

Fleet management should therefore be treated as network operations, not merely aircraft utilization. WSDOT's electric-aircraft planning found that a single electrified airport has limited utility beyond pilot training, while a useful regional air mobility system requires a network of airports aligned with passenger demand, cargo needs, energy infrastructure, and transportation corridors. This network framing also supports multimodal logistics: RAM can connect cargo distribution centers to rail, road, sea, trucks, vans, and smaller UAS rather than simply competing with those modes.⁸

Traffic operations become harder as flights move away from airport-centric patterns into low-altitude urban and regional corridors. Vascik notes that changes to controlled airspace design can require millions of dollars and years of process, making it likely that UAS and ODM aviation will initially operate within status quo Class A-E definitions. Below 1,200 ft AGL, much airspace outside airport areas is uncontrolled Class G, creating an opportunity for new low-altitude ATM designs, but early choices may become codified and difficult to change. Competing concepts illustrate the strategic choice: segregated altitude bands may constrain future ODM operations, while service-provider models could allow UAS and ODM aircraft to share airspace with negotiated priority.⁶

Operational management must also account for community impacts because high-frequency, low-altitude VTOL operations may create noise and visual pollution concerns, especially during takeoff and landing. Miami-Dade's UAM roadmap states that future ATM must cover ATC, traffic flow management, advisories, communication, navigation, and surveillance, while near-term operations are expected to use existing air traffic services and NAS infrastructure. The practical implication is that early AAM operations should be constrained, observable, and procedurally conservative, while the long-term system must be designed for scalable fleet dispatch, resilient airspace operations, and public acceptance.³⁵

Airspace system design and implementation

Airspace design is the scaling bottleneck that turns UAM from a vehicle deployment problem into a managed-network problem. The market architecture already treats "airspace & traffic infrastructure" as a core infrastructure category alongside ground and charging infrastructure, not as an afterthought. That framing is consistent with the broader UAM concept: city-scale flights require integration into existing air traffic management so vertical airspace can function as a dynamic mobility layer rather than isolated routes.¹³¹²

Near-term implementation should remain conservative. Piloted operations currently dominate because they fit existing regulatory frameworks and safety requirements, and early commercial deployments use human oversight to validate operations, build trust, and maintain compliance before any broader transition toward autonomy. This also matches the FAA-oriented concept noted in market sources: FAA UAM ConOps v2.0 emphasizes operations in visual and instrument meteorological conditions while retaining FAA authority over the National Airspace System.¹³¹²

The implementation path is likely to progress through pilots, corridors, and digital traffic services. In March 2026, U.S. government approvals for several UAM pilot projects were described as part of an AAM integration program intended to validate operational frameworks and accelerate eVTOL integration into national airspace. Internationally, EASA's U-Space/UTM package supports high-density drone and eVTOL operations through digital services for traffic management, identity verification, and geo-awareness. These models point toward an airspace system in which authorization, surveillance, conflict management, geo-awareness, and operational priority are increasingly software-mediated.¹³¹²

Dynamic management will matter most once traffic density rises. Advanced UTM/ATM systems are described as using AI and machine learning to process real-time weather, traffic, and no-fly-zone data for dynamic path optimization and conflict avoidance. However, this capability depends on a supporting ecosystem: VTOL manufacturers, infrastructure developers, service providers, mobility operators, regulators, and city authorities must coordinate around safety, operational efficiency, and integration with existing transport systems.¹²¹³

Vertiport rules are inseparable from airspace rules. Vertiports and vertistops provide landing, charging, and passenger handling functions, and market sources identify them as essential to safe, efficient, scalable operations. Regulatory efforts therefore need to cover not only flight corridors but also urban design, energy networks, public safety protocols, noise reduction, cybersecurity, and scalability. The practical design challenge is to certify routes, procedures, data services, and ground interfaces as one operating system.¹³¹²

For this study, the implication is direct: early UAM airspace should be designed as a constrained, observable operating environment, with piloted service, defined routes, limited density, and explicit integration with existing ATC. Long-term scalability depends on performance-based standards, interoperable UTM/ATM services, and harmonized rules that let city-level deployments expand without fragmenting the airspace system.¹²

Low-altitude ATC, aircraft separation, and controlled airspace utilization

Low-altitude AAM capacity is governed less by cruise performance than by how aircraft enter, traverse, and exit already managed urban airspace. The Los Angeles basin illustrates the constraint: within Los Angeles County, the reviewed system included 27 airports, 138 registered heliports, many emergency helicopter landing facilities, LAX Class B airspace, federal airways, VFR and helicopter routes, Class C/D/E airspaces, defense areas, protected flight areas, prohibited airspace, and other competing demands. Controllers also separate helicopters by altitude from faster fixed-wing traffic, which limits the freedom to insert dense ODM traffic as a simple additional layer.⁶

Controlled airspace imposes distinct entry and equipage requirements. In Class B, entry requires ATC clearance and compliance with assigned routing; aircraft under 19,000 lb are separated by at least 500 ft vertically, target resolution, or visual separation, and must carry Mode C transponders and two-way radios. Class C and Class D airspace require two-way radio communication before entry, while Class C requires Mode C transponders; speed limits also apply near primary airports and below 10,000 ft MSL. The Mode C veil around major airports extends 30 nautical miles from Los Angeles, San Diego, and Miramar airports from the surface to 10,000 ft MSL, requiring Mode C transponders and, from 2020, ADS-B for vehicles operating in the veil.⁶

This makes early AAM integration a procedural problem before it becomes an automation problem. One-off ODM missions can use the existing ATC system by communicating with controllers, but rising operation counts would eventually burden controller workload. The key scaling variable is separation: low-altitude throughput depends on both separation minima and controller workload, while VFR rules place separation responsibility on pilots rather than ATC. Improved positioning and sensing have already reduced uncertainty: GPS reduced position uncertainty relative to radar, NextGen reduced ATC separation standards, and onboard systems such as ADS-B have supported both lower separation requirements and safer VFR self-separation.⁶

Dynamic, fine-scale airspace allocation is the main proposed bridge between today's controlled-airspace operations and high-density ODM service. The concept would allocate dynamic fine-scale reserved airspaces, or DFRAs, within surface-level controlled airspace to reduce controller workload while preserving separation assurance. A plausible evolution begins with conventional ATC handling low-volume ODM flights, then moves to DFRAs with VFR self-separation and dedicated communications, then to electronic VFR using advanced technologies to reduce separation minima, and finally to an airspace service provider managing network planning, 4D trajectory optimization, and separation assurance at extreme densities.⁶

Vertiport capacity and airspace capacity must be treated together. A vertipad-capacity model that includes descent, charging, and liftoff time explicitly excludes any additional "safety intervals" that a future traffic-management system might require, meaning pad throughput estimates can overstate real departure capacity if sequencing or separation buffers are later imposed. Charging can also bind operations: the modeled scenario used 30 minutes of battery charge time, with disembarking, cleaning, and boarding assumed comparable to charge time. Thus, controlled-airspace utilization is not only an en-route question; it includes arrival sequencing, pad occupancy, charging duration, and departure release.¹⁴

The practical implication is a staged operating doctrine. Near-term AAM should use existing ATC, defined routes, conservative separation, and constrained volumes where controller workload remains manageable. Medium-term growth requires digital corridors, real-time deconfliction, and mixed radio/datalink coordination, consistent with reported UAM efforts around fixed routes, predefined altitudes, corridor operations, and future AAM traffic-management systems. Far-term density requires service-provider-based traffic management with reliable surveillance, communications, trajectory planning, and separation assurance, because unmanaged VFR self-separation is expected to face throughput and safety concerns as aircraft density rises.³⁸⁶

Low-altitude ATC is therefore the central scalability gate for urban AAM. Aircraft noise may remain the strongest near-term public constraint, but airspace throughput becomes decisive as networks densify: if separation standards, controller workload, and vertiport sequencing are not jointly managed, higher vehicle production or stronger demand will not translate into usable network capacity.⁶

Navigable airspace and low-altitude operating limits

Low-altitude AAM cannot be planned as a simple straight-line substitute for road travel. The relevant operating envelope includes FAA terminal-area charts, ATC entrance requirements, airspeed limits below 10,000 ft MSL, controlled-airspace boundaries, flight minimums, low-altitude separation standards, and surface-level controlled airspaces. In practice, each origin-destination pair must be tested against these constraints, not only against distance and ground congestion.⁶

This is especially important for point-to-point ODM missions, which are less temporally and directionally predictable than commuter flows and may run from one location to many possible destinations throughout the day. The Rancho Palos Verdes to Torrance Memorial Hospital mission illustrates the operational tradeoff. It is only 8.5 miles, with ground travel as low as 16 minutes and as high as 28 minutes during peak congestion, yet medical urgency may still create high time value despite lower trip frequency. For such short missions, low-altitude routing, approach/departure procedures, airspace entry requirements, and vertiport access can dominate the business case.⁶

The study's use of randomly selected origin and destination points is therefore methodologically important: those missions were intended to reveal constraints or opportunities not visible in high-value market cases. A viable network must handle both premium routes and less curated OD patterns, because navigable airspace availability is uneven across geography, altitude, and operating context.⁶

Low-altitude operations also face environmental limits that are distinct from conventional aviation. Urban UAM flights are exposed to micro-weather, including wind-speed gradients, urban updrafts and downdrafts, building wake shear, small-scale turbulence, vortex shedding, and localized icing. These conditions change over smaller spatial and temporal scales than traditional aviation weather, requiring improved stationary sensing, onboard sensing, micro-weather forecasting, and decision-support tools. As a result, "navigable" airspace is not merely legally available airspace; it is airspace that can be used safely, predictably, and repeatedly under local operating limits.¹



Exhibit. Graph of EHang 216 Source: EHang Research. Source: S0075.

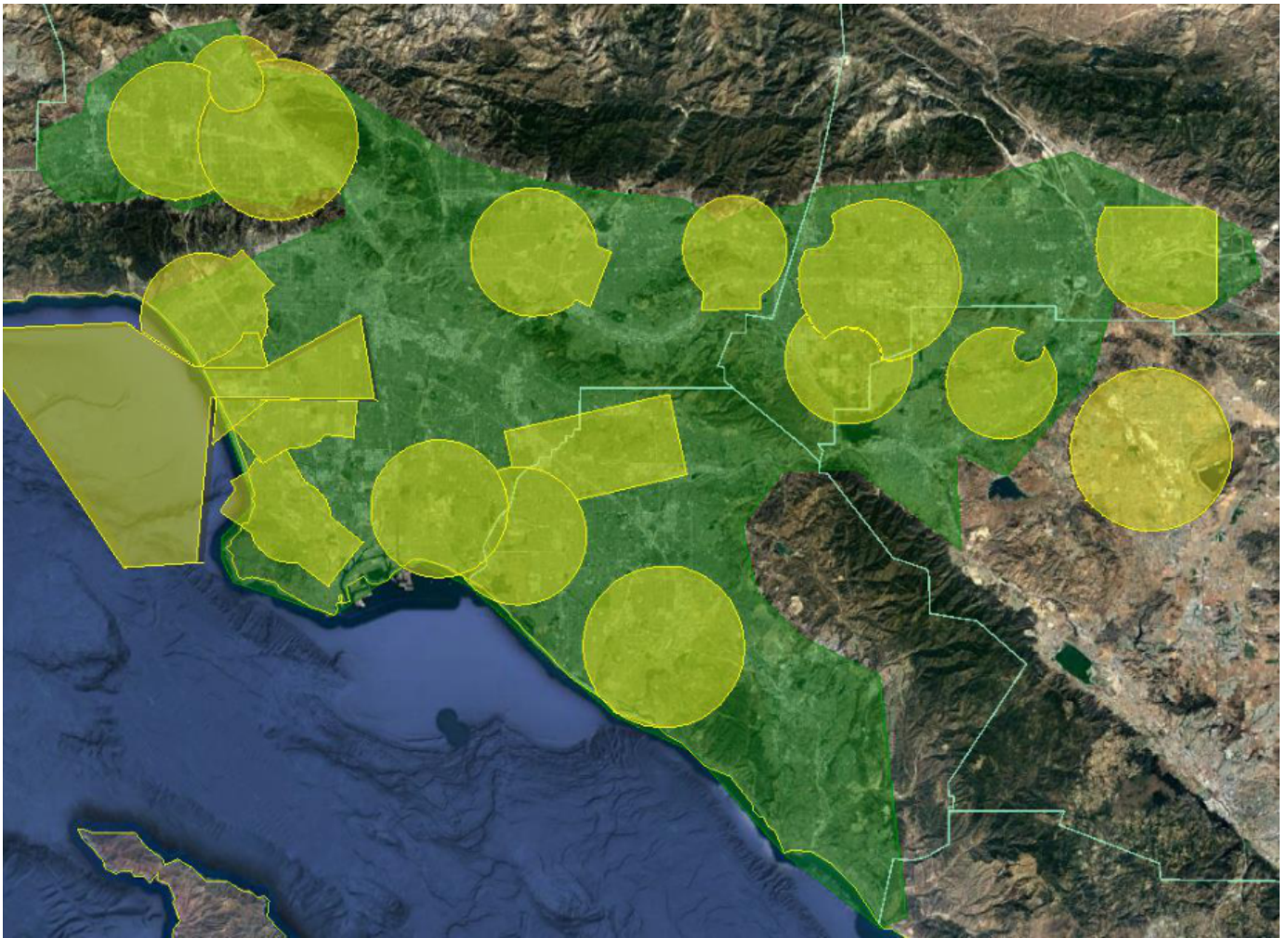


Exhibit. Figure 83. Surface-level controlled airspaces (yellow) cover 43% of the densely populated areas (green) in the L. Source: S0011.

Landowner rights, aviation easements, trespass, nuisance, and privacy

Low-altitude AAM adds a property-law layer to the operational constraints discussed above. The core unresolved questions are who owns or regulates low-altitude airspace, and how landowner rights are balanced against a public right to fly and use the National Airspace; the specific tort pathways include trespass, nuisance, voyeurism, and invasion of privacy.⁶

One proposed resolution is a private-airspace demarcation altitude. Below that boundary, aircraft would need landowner permission or risk trespass liability; above it, operators would gain stronger protection from trespass and privacy claims based on ordinary naked-eye observation. But that line would not eliminate other exposure. The same source emphasizes that flights above the demarcation could still face nuisance, takings, or privacy claims involving advanced imaging devices.⁶

The more practical legal model is therefore not a single altitude rule but a “continuum of acceptability.” As altitude increases, the public interest in air sovereignty becomes stronger and the landowner’s interest weaker, so a landowner would need to show more severe harm at higher altitudes. This matters for AAM because disturbance footprint varies by vehicle type: small UAS may operate closer to the demarcation if their noise, light, and vibration are limited, while electric ODM aircraft may need higher routine altitudes except during takeoff and landing.⁶

Aviation easements are the clearest existing mechanism for reconciling these interests near frequent takeoff and landing sites. Airports use them to secure airspace volumes over neighboring properties for safe aircraft passage, restrict incompatible obstacles, compensate landowners, and reduce liability from trespass, nuisance, and privacy claims. As UAS and ODM takeoff-and-landing areas become more common and congested in communities, the same mechanism may become more widely used.⁶

This legal exposure also links directly to public acceptance. NASA-linked survey evidence found consumer concerns around safety, privacy, job security, environmental effects, noise, and visual pollution, with last-mile delivery

respondents specifically worried about privacy breaches from vehicle cameras. For AAM deployment, property-rights strategy should therefore be treated as part of route design and vertiport siting, not as a late-stage legal cleanup.¹⁶

Regulatory pathways, risk management, and timeline acceleration

Regulation is not a final permitting step after aircraft readiness; it is a pacing constraint for the whole AAM system. Evidence from the UK review indicates that some AAM regulatory actions may extend more than 10 years, while airspace change processes can take two to eight years depending on the change required. This creates a timing mismatch: vehicles may be ready for initial operations before the airspace, operating rules, infrastructure standards, pilot-licensing rules, and certification frameworks are mature enough to support them.¹⁷

The practical pathway is phased. NASA's operational concept expects the initial state to use certified, human-piloted vehicles with supporting automation, existing heliports, expanded helicopter routes and procedures, limited traffic volumes, tightly controlled applications, and ongoing development of UAM traffic-management capabilities. The intermediate state then shifts toward semi-autonomous vehicle and ATM subsystems, expanded infrastructure, enhanced CNSI, reduced-visibility capabilities, and codified regulatory and liability standards. This supports a risk-managed sequence: prove narrow operations first, gather operational data, then expand only as automation, airspace procedures, community acceptance, and liability rules mature.³⁰

Risk management should be mission-specific rather than generic. Vascik's methodology evaluated ODM operations through technical feasibility, regulatory compliance, community acceptance, ATC interfaces, ground infrastructure, ground transportation, and environmental impact lenses, with ConOps covering customer access, staging, boarding, flight-path planning, ATC interaction, charging, and other operational steps. That structure is important because point-to-point ODM missions are less predictable than commuter flows: they can occur throughout the day, in varied directions, and between diverse origin-destination points. The Rancho Palos Verdes to Torrance Memorial Hospital case illustrates why regulatory screening cannot rely on distance alone: the trip is only 8.5 miles and 16-28 minutes by car, but medical time value may justify air service even though such missions are likely less frequent than commuter or ordinary point-to-point trips.⁶

Timeline acceleration should therefore focus on temporary, evidence-generating permissions rather than premature full-scale liberalization. The UK evidence review proposes rolling airspace trials so initial AAM operations can proceed while longer airspace-change processes continue, with benefits including greater manufacturer and investor confidence, public exposure to AAM services, and operational data for regulators. It also identifies greater regulator resourcing and talent retention as a way to streamline certification and licensing, while noting that this only helps if industry can meet the safety standards and no institutional inefficiencies prevent process improvement.¹⁷

The regulatory risk register should prioritize three near-term gaps: certification and operating-rule clarity, airspace integration, and international standards alignment. Industry interviewees cited uncertainty over when standards will be adopted and what principles will govern infrastructure, airspace integration, air operations, vehicle design, battery charging, and pilot licensing. Airspace integration remains a safety and capacity concern because AAM must coexist with other aircraft, weather constraints, ATC workload, autonomous operations, and potentially dynamic restrictions in lower airspace. Global harmonization also matters because divergent vertiport and certification standards could delay cross-border deployment or raise production costs.¹⁷

The fastest credible path is not to compress every approval into one launch date. It is to stage approval around constrained corridors, defined missions, low traffic volumes, conservative separation, transparent public exposure, and regulator-observed trials, then convert demonstrated evidence into durable rules. That approach aligns the operational maturity curve with the regulatory one: initial piloted service, measured expansion, semi-autonomous capability, and only later high-density scaled AAM.^{30 17}

Economic, Environmental, and Social Impacts

CapEx, OpEx, revenue, and operating hours

AAM economics should be modeled as facility-and-network economics, not only aircraft economics. In Orlando's Geico Garage case, VHB treated the downtown vertiport as one node in a 14-vertiport Florida network and used passenger revenue, capital cost, employment, and compensation as IMPLAN inputs. The retrofit CapEx assumption

was \$30 million, covering charging infrastructure, passenger and crew facilities, MRO facilities, vehicle gates, and contingency for structural reinforcement because the garage roof was not originally designed for vertiport structures. Earlier workshop inputs also showed how sensitive the estimate is to definition: Lilium's surface vertiport estimate was \$10 million, while an "urban garage infrastructure" estimate was \$20 million and the later Geico Garage retrofit assumption was \$30 million.⁴

The same case shows that public-sector economics include opportunity cost, not just construction spend. Converting the garage roof was estimated to displace \$600,000 to \$1 million in annual parking revenue, while a potential ground lease to an AAM operator was estimated at \$2.1 million per year, enough to offset that parking loss in the scenario, though the lease was not included in the IMPLAN input set. This matters because vertiport viability depends on property terms, structural retrofit risk, revenue sharing, and lease allocation as much as on flight demand.⁴

OpEx assumptions remain comparatively immature. Lilium did not provide facility-level OpEx estimates, but identified ground lease, staffing, utilities, maintenance, and taxes as major categories; investor-facing cost data instead emphasized eVTOL operating cost, cited in the Orlando material as \$2.25 per mile. Another Orlando input table split costs into 50% fixed and 50% variable, with fixed cost categories including infrastructure, pilots, data cost, and air traffic fees, and variable categories including maintenance and spares, tax, depreciation, customer acquisition, battery replacement, and energy. The same forecast applied 2.5% annual inflation after the 2023 base year and assumed construction completed in Year 1, with full vertiport utilization reached in Year 4, 2026.⁴

Operating hours should therefore be treated as a constrained utilization scenario rather than a free revenue lever. The Orlando/Lilium material assumes shift-based staffing and identifies 143 Lake Nona jobs, including 60 pilots or flight instructors, 55 ground operations roles, and 8 customer service or administrative roles. But utilization can be capped by infrastructure, ATC, and community constraints: viable UAM infrastructure requires vertiports, electricity, navigational aids, and robust air traffic management systems, while aircraft noise was identified as perhaps the most significant near-term ODM aviation constraint and may affect TOLA location, aircraft type, operations, and demand. High operating-hour assumptions are only credible if the model also carries these constraints into CapEx, OpEx, revenue, and schedule phasing.⁴³⁵⁶

IMPLAN input assumptions and economic impact assessment

IMPLAN is useful for AAM only when the aviation concept is translated into explicit facility, network, revenue, cost, and employment assumptions. In the Orlando Geico Garage case, the modeled vertiport was assumed to operate within a 14-vertiport Florida network, and IMPLAN inputs included passenger revenue, capital cost, employee count, and employee compensation. The base retrofit CapEx was set at \$30 million, covering electric charging, passenger and crew facilities, MRO facilities, vehicle gates, and structural-reinforcement contingency. Because the garage roof was not originally designed for vertiport structures, the source notes that a detailed structural assessment would be needed to understand retrofit cost risk.⁴

The Orlando analysis separated construction and operations impacts, treating construction as a one-time impact and applying multi-region input-output analysis so Orlando, Orange County, and Florida results would not be double-counted. It also excluded eVTOL vehicle CapEx because the aircraft were assumed to be manufactured in Germany, illustrating how local-content assumptions determine what economic activity is captured in the study area. Operating assumptions included a phased ramp to full utilization by year four, a 2.5% inflation factor after the 2023 base year, and cost categories split between fixed and variable operating costs.⁴

The model's outputs should be read as conditional economic-impact estimates, not standalone proof of feasibility. During operations, Orlando estimated total output of about \$52 million, equal to first-full-year annual passenger revenue, and annual value added of \$35.9 million across Orlando, Orange County, and Florida. The same results table reported 202 total jobs and 10-year tax revenues of \$2.8 million for Orlando, \$9.6 million for Orange County, \$29.3 million for Florida, and \$24.6 million federally. Employment impacts include direct jobs at the vertiport, indirect jobs in AAM-supporting supply-chain industries, and induced jobs from subsequent spending across the economy.⁴

The Orlando case also shows why public-sector fiscal analysis must sit beside IMPLAN outputs. Converting the garage roof could displace roughly \$600,000 to \$1 million in annual parking revenue and require renegotiation of an Orlando Magic parking lease, while a possible \$2.1 million annual ground lease to an AAM operator was identified but not included in the IMPLAN model. This distinction matters: IMPLAN captures modeled economic circulation, but

the sponsoring city still needs a project-level pro forma that accounts for foregone revenue, lease structure, cost sharing, and retained risk.⁴

WSDOT's economic-impact framework provides the broader interpretation. Direct impacts are initial aviation-related activity, while multiplier impacts include supplier sales and worker income re-spending; reported measures include jobs, labor income, value added, and business revenues. Value added is the relevant GDP/GSP contribution because it measures business revenues net of purchased intermediate goods and services. For AAM, that means economic-impact claims are strongest when tied to concrete networks of airports, charging, passenger/cargo demand, and multimodal logistics rather than a single isolated vertiport. WSDOT explicitly notes that one electrified airport has limited utility beyond pilot training and that RAM value depends on airport networks aligned with passenger, cargo, energy, and corridor needs.⁸

GDP, employment, taxes, supply chain, and job creation

Economic impact analysis is useful here only if it is tied to specific business-case inputs, not generic AAM optimism. The Orlando Geico Garage study did this by translating a conceptual vertiport into IMPLAN inputs: passenger revenue, capital cost, employee count, and compensation within a 14-vertiport Florida network. Its retrofit assumption was \$30 million, covering charging infrastructure, passenger and crew facilities, MRO facilities, vehicle gates, and structural reinforcement contingency.⁴

The resulting GDP impact table reported \$35.88 million in total GDP impact, consisting of \$19.07 million direct, \$9.49 million indirect, and \$7.31 million induced GDP across Orlando, Orange County, and Florida. A separate Orlando table reported \$7.40 million in total GDP, with \$5.28 million direct, \$1.39 million indirect, and \$0.73 million induced. These outputs should be read as scenario results from stated inputs, because IMPLAN estimates economy-wide spending patterns and depends on the assumed business activity entered into the model.⁴

The Oklahoma analysis uses the same core logic at broader scale: NEXA combined business-case forecasts with IMPLAN for AAM aircraft other than drones across six Oklahoma regions from 2024 to 2045. Its model links revenue, cost, CapEx, and OpEx to direct, indirect, induced, and catalytic impacts, with consolidated OpEx, CapEx, and revenue generated across infrastructure, UTM, aircraft costs, and operator revenue supply chains. In this framing, direct effects come from the AAM activity itself, indirect effects from supplier demand, and induced effects from household spending by employees in the direct and supply-chain businesses.³¹

Job creation is therefore not limited to pilots or vertiport staff. Orlando's Lake Nona assumptions anticipated 143 jobs, including 60 pilots or flight instructors, 55 ground operations roles, and 8 customer service or administrative roles. The same workshop material identified the key vertiport supply chain as ground infrastructure, AAM operators, AAM traffic management, and AAM manufacturing. Oklahoma's report similarly treats new jobs as the channel through which AAM supports businesses, GDP, and tax revenues for public services.^{4,31}

Tax and public-revenue conclusions are more fragile than GDP headlines because they depend on local deal structure. Orlando estimated \$600,000 to \$1 million in annual lost parking revenue from repurposing 300 roof-deck spaces, while a potential \$2.1 million annual ground lease could offset that loss, though the lease was not included in the IMPLAN inputs. For public agencies, the implication is straightforward: AAM economic-development claims should be evaluated net of displaced uses, lease terms, infrastructure risk, staffing assumptions, and the maturity of the supply chain.⁴

Life-cycle analysis and externality assumptions

AAM environmental accounting should use life-cycle boundaries, not flight-only energy. The relevant LCA steps are scope definition, inventory analysis, human-health impact assessment, and uncertainty testing through sensitivity analysis. For eVTOLs, the core LCA categories are energy consumed, environmental externalities, and operating costs, each split between manufacturing and operation.²²

The modeled eVTOL boundary includes vehicle and vertiport manufacturing emissions, vertiport equipment, electricity consumption, platform rental, maintenance labor, and operating emissions from power plants used to recharge batteries and power ports. This makes electricity mix a first-order assumption: the cited Texas ERCOT CO₂e rate declined from 239 kg/MWh in 2016 to 223 kg/MWh in 2018 and 195 kg/MWh in 2021, while nuclear, wind, and solar generation are treated as having no operating emissions.²²

Battery assumptions are especially sensitive. One eVTOL study assumes batteries cost \$700/kWh, provide 230 Wh/kg, last 2,000 cycles, and require charging from a 20% reserve; slow charging could materially restrict utilization, while a one-hour charge for a 100-kWh battery is used as a more practical assumption. Battery production is also a major embodied-emissions source: cited estimates range from 38 to 356 kg CO₂e/kWh of capacity, and the study assumes 150 kg CO₂e/kWh for the battery production phase.²²

Structural materials and propulsion add further embodied impacts. The vehicle body is assumed to use material slightly heavier than carbon fiber, with material-plus-assembly cost based on a Vahana trade-study assumption of \$100/lb and tooling at \$10,000/m³. For structural components, the study assumes 35 kg CO₂e per kg of material to cover manufacturing, shaping, and stamping. Because direct rotor-motor manufacturing emissions data were unavailable, motor life-cycle emissions are estimated from electric-vehicle traction-motor evidence and applied as a conservative power-based assumption.²²

Vertiports should be counted as part of the environmental system, not treated as neutral access points. Infrastructure energy and emissions matter across modes, and one cited estimate assigns air-mode taxiway construction 95 MJ and 6.8 kg CO₂e per square foot, compared with a Texas-reported 13 kg CO₂e per square foot. Operational assumptions also include lighting, charging stations, passenger areas, maintenance access, and facility space; the study assumes each eVTOL needs a facility area 20% larger than its footprint, with surrounding operations and passenger space ten times larger.²²

The externality set remains incomplete. The eVTOL LCA explicitly covers battery and vehicle-parts disposal, noise, power-based emissions, and manufacturing and maintenance emissions, but excludes some safety and passenger-harm concerns because of limited data. This limitation matters for feasibility interpretation: LCA can compare energy and emissions pathways, but it cannot by itself settle community acceptance, safety, privacy, or nuisance impacts.²²

Noise, visual disruption, and ground population exposure

Noise is not a secondary externality for AAM; it is part of community integration, land use, environmental policy, and local restriction design. NASA's market study identifies "community integration" as one of the five regulatory categories that must be addressed for last-mile delivery, air metro, and air taxi operations, alongside air traffic, vehicle production, airspace design, and individual vehicle operations. Booz Allen's NASA briefing similarly places privacy and noise among societal barriers, while noting that UAM legal challenges extend into environmental policy, public use, land use, and local restrictions.²¹²

Electric and hybrid propulsion may reduce noise and pollution relative to conventional aviation concepts, but that advantage does not eliminate exposure concerns. The relevant planning question is not only whether each aircraft is quieter, but where operations concentrate, how often flights occur, which populations sit below approach and departure paths, and whether visual annoyance, safety concerns, and public-health impacts accumulate with scale. Illinois' AAM framework explicitly treats noise pollution, visual annoyance, emissions, safety concerns, public health, and energy consumption as negative environmental aspects requiring assessment.³⁰²⁶

Ground exposure is shaped by infrastructure and network design. Miami-Dade's UAM roadmap notes that ground-infrastructure needs depend on vertiport location, use case, aircraft technology, and implementation strategy, and recommends evaluating specific locations, VTOL types, charging or fueling needs, vehicle size, noise profile, number of operations, passenger activity, and utility availability before defining integration needs. This makes noise and visual disruption a site-specific system output, not a generic aircraft attribute.³⁵

Operational modeling should also account for non-revenue and repositioning flights. In the Stanford Hopper study, the fleet assignment problem produced a sharp tradeoff between the number of aircraft owned and the number of repositioning flights, with the minimum-fleet case described as the worst case for air traffic because it relies on many repositioning flights. For community exposure, those extra movements matter because residents experience operations, not just paid passenger trips.³²

The practical implication is that AAM environmental review should be corridor- and facility-based. Early analysis should map expected approach paths, departure paths, holding or sequencing areas, repositioning movements, operating hours, nearby sensitive receptors, and affected ground populations. Public acceptance research is also needed because self-reported surveys, focus groups, stakeholder interviews, statistical analysis, and GIS analysis

can help anticipate how communities may respond to emerging transportation technologies and inform proactive policy that maximizes benefits while minimizing adverse effects.²

Social equity, environmental justice, and underserved communities

AAM's equity case is conditional, not automatic. Evidence reviews describe possible benefits for disabled users, underserved communities, regional connectivity, healthcare access, and time-critical medical delivery, but also warn that these outcomes depend on infrastructure design, affordability, physical accessibility, user requirements, and links to existing transport systems. Public participants in UK research specifically raised concerns that future-flight technologies could benefit groups unequally, and identified affordability, accessibility, connection to existing infrastructure, and public management as ways to reduce discrimination by income or geography.¹⁷

The early-market evidence points in the opposite direction unless policy intervenes. AAM is expected to tap into large cities with existing transportation hubs, which improves reach in absolute population terms but can also concentrate service in high-cost markets where early users are high earners or people with high willingness to pay. In the Los Angeles case study, some early reference missions were explicitly selected around high-income communities, while the Antelope Valley mega-commute market was described as less affluent and therefore likely to have lower market penetration despite a larger addressable market of 951 individuals. That distinction matters: equity-relevant demand is not just where congestion is severe, but where willingness to pay, access to takeoff and landing areas, and trip purpose align.¹⁷⁶

Environmental justice requires the same geographic specificity. Vascik's systems-level method evaluated reference missions through social, environmental, regulatory, legal, and technical lenses, including ground population density, infrastructure availability, and airspace interaction. The thesis also explicitly included social equity and environmental justice as externality categories in conceptual design. For AAM planning, this means origin-destination analysis should not only locate profitable trips; it should map who receives time savings, who bears noise and visual exposure, and who is bypassed by vertiport siting.⁶

Reference missions offer a practical screening method. The Los Angeles study selected origin and destination points, validated commute markets with Census LODES data, used inverse laborshed mapping to identify where residents commute, and then checked ground travel-time profiles with Google Maps predictions. The same workflow can be adapted for equity analysis by testing whether underserved communities gain access to jobs, healthcare, education, or regional services, rather than simply identifying affluent commuter corridors. Randomly selected missions were also added to reduce selection bias and reveal constraints or opportunities not present in researcher-selected high-value markets.⁶

Policy design will determine whether AAM becomes a premium bypass or a public-access mobility layer. The evidence review notes that Canada's AAM consortium proposed subsidizing 15% of flights at no or low cost to low-income residents, while finding little evidence that other urban centers planned similar subsidies. Without affordability mechanisms, accessible infrastructure, and deliberate service obligations, early AAM is more likely to serve narrow high-value users than underserved communities.¹⁷

Project Prioritization and Funding

AAM hotspot mapping and data engagement

AAM hotspot mapping should begin with use-case segmentation, not a single generic "UAM demand" layer. The available market sources divide demand by vehicle type, range, operation mode, end user, infrastructure, and geography, including air taxis, air metro, air ambulance, last-mile delivery, intercity and intracity service, piloted/autonomous/hybrid operations, ridesharing companies, scheduled operators, e-commerce companies, hospitals and medical agencies, and private operators. A second segmentation scheme separates passenger transport, cargo and logistics, medical and disaster response, private use, ground and charging infrastructure, airspace and traffic infrastructure, short range under 50 km, medium range of 50-200 km, and long range above 200 km. These categories imply that hotspot maps should be layered by mission type and customer class rather than reduced to population density or congestion alone.¹¹³

The strongest near-term mapping candidates are locations where multiple segment signals overlap. Market sources identify passenger transport, air taxi services, short-range operations, airport connectivity, commercial fleet operat-

ors, vertiports/vertistops, and piloted operations as leading early-market categories. Fortune's segmentation also points to ridesharing companies, scheduled operators, e-commerce companies, hospitals and medical agencies, and private operators as distinct end-user groups, with hospitals and medical agencies holding a significant share due to emergency medical-care demand and rural-treatment needs during the pandemic. A practical hotspot screen should therefore test whether a candidate place can support more than one use case: premium passenger trips, airport shuttle service, medical transport, cargo/logistics, or emergency response.^{[131112](#)}

Forecasts are useful for scenario design but should not be treated as ground truth for local siting. The same market category is reported with materially different baselines and growth rates: MarketsandMarkets estimates the UAM market at USD 1.62 billion in 2025 and USD 16.27 billion in 2035, while ResearchAndMarkets reports USD 12.3 billion in 2025 and USD 34.38 billion by 2034. Regional shares also differ: MarketsandMarkets reports North America at 26.8% in 2026, Fortune reports North America at 40.42% of global revenue in 2025, and Roots reports North America at 41.90%. For hotspot mapping, these differences argue for using market reports as boundary conditions and segment taxonomies, while relying on local trip, infrastructure, stakeholder, and constraint data for project prioritization.^{[13391112](#)}

Data engagement should follow a triangulated workflow. One market study describes a four-step process: secondary research on eVTOL platforms, infrastructure development, airspace management systems, operational models, and end-user adoption; primary interviews with OEMs, infrastructure providers, mobility operators, regulators, and technology experts; analysis of pilot programs, certification progress, investment trends, and partnerships; and segmentation with data triangulation across platform, infrastructure, application, operation mode, range, end user, and architecture. For public-sector AAM planning, this translates into a local engagement program with airports, city transportation agencies, utilities, emergency medical stakeholders, hospitals, freight and logistics users, prospective operators, regulators, and community representatives.^{[13](#)}

Hotspot outputs should therefore be decision layers, not just heat maps. At minimum, each candidate area should be scored by use case, range band, likely operator/end user, infrastructure need, airspace/traffic-management exposure, charging requirement, and stakeholder readiness. This is consistent with market evidence that deployment is shaped by infrastructure limitations, high initial costs, airspace integration complexity, battery limitations, certification complexity, and evolving regulation. The purpose of hotspot mapping is not to prove that AAM demand exists everywhere; it is to identify where a specific AAM mission has enough demand, infrastructure feasibility, operational compatibility, and stakeholder support to justify deeper business-case analysis.^{[13](#)}

Potential project areas and scoring criteria

Project selection should start from use cases, then test each candidate against geography, infrastructure, demand, cost, risk, and public value. PennDOT's AAM feasibility work used seven representative business cases: airport shuttle, cargo and freight delivery, corporate aviation, emergency services, on-demand air taxi, regional air mobility, and specialized services such as tourism, VIP transport, search and rescue, security, and agriculture. Its project identification framework then scored candidate areas using seven criteria: implementation feasibility, cost, risk, stakeholder support, sustainability, demand, and alignment with federal and state goals.^{[20](#)}

The strongest project areas are those where operational need is specific and near-term constraints are manageable. PennDOT's highest-scoring examples were corridor freight support, rapid incident response, urban-core-to-urban-core personal transport, rural emergency services, airport shuttle, and remote medical delivery. This ordering is consistent with the broader evidence: high-value early markets include long-distance intercity trips, medical or deadline-sensitive trips, congested airport/event routes, business travel, and recreation trips where ground travel is slow or unreliable.^{[206](#)}

Scoring should penalize projects that depend on diffuse, low-frequency, or poorly located demand. In the Los Angeles reference missions, Malibu to Century City and San Bernardino to Glendale were promising but small-volume commute markets, with addressable markets of 43 and 85 individuals, respectively; adding broader destination options increased the potential market to more than 800 individuals. The Antelope Valley to Los Angeles City Center route had a larger addressable market of 951 individuals, but lower affluence implied lower expected market penetration. These examples suggest that "demand" should combine trip volume, willingness to pay, congestion relief, destination breadth, and infrastructure access rather than rely on any single indicator.^{[6](#)}

Medical and emergency projects deserve separate treatment. The Rancho Palos Verdes to Torrance Memorial Hospital mission did not show strong one-way time savings because nearby urgent-care travel times were short and not

heavily congestion-driven. However, the same source notes that medical ODM could be more compelling for access to specialized doctors, on-site landings, ambulance substitution, or round-trip emergency response. A scoring framework should therefore distinguish routine patient transport from emergency response, organ transport, rural access, and specialized-care missions.⁶

Finally, financial scoring must include site-specific opportunity costs, not only aircraft or vertiport costs. Orlando's conceptual downtown vertiport analysis assumed \$30 million to retrofit the Geico Garage, including charging, passenger and crew facilities, maintenance facilities, gates, structural contingency, and reinforcement risk; it also identified \$600,000 to \$1 million in annual lost parking revenue from repurposing roof-deck spaces. A credible project score should therefore require explicit assumptions for capital cost, operating cost, revenue, lost existing use, stakeholder support, and funding availability before a candidate is treated as investment-ready.⁴²⁰

Strategic investment priority projects

Strategic investment should move from broad AAM interest to a sequenced portfolio of projects tied to defined use cases, infrastructure classes, and operators. Market segmentation consistently separates vehicle types such as air taxis, air metro, air ambulance, and last-mile delivery; range categories such as intercity and intracity; operating modes such as piloted, autonomous, and hybrid; and end users such as ridesharing companies, scheduled operators, e-commerce firms, hospitals, medical agencies, and private operators. A parallel market taxonomy adds infrastructure categories, including air traffic control systems, charging stations, maintenance, repair and overhaul, and vertiports. These categories imply that priority projects should be evaluated as network investments, not isolated aircraft demonstrations.¹¹¹²

The first priority is an airport-based regional network pilot using existing airports as operating nodes. Jiang et al. propose a UAM model that leverages existing regional airports and co-optimizes demand, fleet operations, and ground-transport interactions through LPSim. In the San Francisco Bay Area case, the framework evaluated a scenario in which 21 regional airports were fully dedicated to UAM operations. This type of project is investment-ready because it forces early decisions on fleet sizing, charging windows, repositioning, airport access, and ground integration before expensive infrastructure is overbuilt.²⁵

The second priority is a fleet-optimization and scheduling project. The cited methodology first assumes an unconstrained fleet, assigns the most suitable aircraft by range, runway requirement, and party size, and then converts qualifying UAM trips into flight tasks with origin, destination, operating time, and aircraft type. It then uses a time-expanded minimum-cost network-flow model to calculate the minimum fleet required to serve those tasks. Transition feasibility depends on whether one aircraft can complete sequential tasks after repositioning flight time and required charging time. This makes fleet optimization a core capital discipline: the objective is not maximum vehicle count, but the smallest fleet that can satisfy demand under operational constraints.²⁵

The third priority is an infrastructure-readiness program focused on charging, vertiports, MRO, and airspace-management interfaces. Market reports identify charging stations, vertiports, maintenance, repair and overhaul, and air traffic control systems as explicit UAM infrastructure segments. Industry sources also note that several companies are pursuing deployment through collaborations and partnerships, while substantial groundwork remains for necessary infrastructure and air traffic management. Public investment should therefore favor shared enabling assets that multiple operators can use, rather than single-operator facilities.¹²¹¹

The fourth priority is a medical and emergency-services corridor portfolio. UAM market segmentation includes air ambulance and medical emergency vehicles as a distinct vehicle class, and hospitals plus medical agencies appear as defined end users. Fortune Business Insights reports that hospitals and medical agencies accounted for a significant share because of increased emergency medical-care demand, including pandemic-era demand for air ambulances serving remote areas. This use case deserves priority where time sensitivity is high and public-benefit justification is stronger than for premium passenger service.¹²¹¹

The fifth priority is selective commercial service testing across air taxi, air shuttle, cargo, and last-mile delivery. Air taxis are a leading commercial segment in multiple market reports: Fortune reports air taxis at 35.05% market share in 2026, while Roots reports air taxis as 63% of the current market by vehicle type. Cargo air vehicles and last-mile delivery vehicles are also distinct market categories, with e-commerce companies identified as an end-user group. These pilots should be structured as measured demonstrations with transparent assumptions, since market forecasts vary widely: Fortune reports a 16.6% CAGR for 2026-2034, while Roots projects 27.85% CAGR for 2026-2040.¹¹¹²

Federal, state, regional, local, private, and industry funding

AAM funding should be treated as layered system finance, not a single aviation capital program. The federally relevant work is the enabling layer: NASA, FAA, other federal agencies, industry, and academia are already described as collaborating on AAM infrastructure, information architecture, software functions, concepts of operation, operations-management tools, and other functional components. Federal support is therefore most defensible where benefits are shared across markets: certification pathways, low-altitude traffic management, communications, safety analysis, and scalable operating concepts.²³

Airspace-management investment is a particularly strong candidate for public funding because it is a network constraint rather than an operator-specific asset. In Los Angeles, Vascik found that large portions of surface-level controlled airspace may be underutilized under current ATC configurations, creating a possible role for dynamic, fine-scale airspace allocation. The proposed DFRA concept would let controllers release predefined airspace volumes for properly equipped aircraft under special flight rules, reducing ATC workload while enabling low-altitude operations. That kind of shared capacity expansion is poorly matched to individual project finance but well matched to federal and regional demonstration funding.⁶

State, regional, and local funding should focus on place-specific assets: vertiports, access links, charging, passenger facilities, structural upgrades, and integration with existing transportation infrastructure. The Orlando Geico Garage case illustrates the scale and specificity of these costs: VHB used a \$30 million retrofit assumption covering charging infrastructure, passenger and crew facilities, MRO facilities, vehicle gates, structural reinforcement contingency, and related vertiport needs. The same case also shows why local governments need site-level revenue analysis; converting the garage roof could displace an estimated \$600,000 to \$1 million in annual parking revenue, while a potential \$2.1 million annual ground lease to an AAM operator could offset that loss.⁴

Public agencies may also contribute land or rights-of-way where AAM infrastructure overlaps with existing transportation assets. NASA-linked VTOL literature identified freeway cloverleaf interiors as candidate vertiport or vertistop locations because freeway noise may reduce incremental sound concerns, interchange locations can provide vehicle access, and such infrastructure may already be publicly owned by transportation agencies. This supports a state or regional funding model in which public entities provide siting, permitting coordination, utility access, and intermodal connections, while private operators finance aircraft and revenue-facing services.²²

Private and industry funding is already part of the AAM development model. Airbus, Boeing, and other corporations are described as devoting money and time to VTOL design and operation, and Airbus' Vahana, Opener BlackFly, and other prototypes had early flight activity by 2018. Orlando's planning work also relied on stakeholder interviews with Lilium and public investment information from other AAM manufacturers and operators to build assumptions for capital costs, operating costs, and revenue projections. This suggests private capital is best directed toward vehicles, operations, leases, customer acquisition, and commercially specific infrastructure, with public capital reserved for shared enabling systems and public-interest conditions.²²⁴

Funding decisions should also require environmental and life-cycle accounting. eVTOL externalities include battery and vehicle-part disposal, noise, power-generation emissions, manufacturing emissions, maintenance emissions, and vertiport construction and operation impacts. Because LCA requires defined boundaries, inventory analysis, health-effect assessment, and uncertainty analysis, funding applications should not rely on flight-only emissions claims. A credible funding package should therefore pair capital support with requirements for utilization reporting, charging data, grid-emissions accounting, noise monitoring, community engagement, and post-deployment performance review.²²

Airport funding competitiveness

Airport AAM funding is most competitive when framed as shared enabling infrastructure, not speculative air-taxi promotion. Market evidence points to vertiports, vertistops, charging, passenger handling, and airspace-management integration as core deployment needs: vertiports/vertistops accounted for 52.2% of the UAM infrastructure segment in 2026, while infrastructure gaps, high deployment costs, certification complexity, airspace integration, traffic management, battery limits, and performance constraints remain stated market barriers. This makes airport projects stronger candidates when they reduce common operating constraints across operators and use cases rather than subsidizing a single vehicle company.¹³

Airports have a defensible role because early UAM activity is expected to concentrate around short-distance routes, airport connectivity, piloted operations, and commercial fleet operators. MarketsandMarkets identifies short-range operations as prevalent because they match current battery capabilities and early airport-connectivity demand, and it states that piloted operations dominate because they fit existing regulatory and safety frameworks. The same source describes vertiports and vertistops as facilities for landing, charging, and passenger handling, with city-level initiatives and pilot projects accelerating deployment. For airports, this implies a competitive funding narrative around readiness: electrical capacity, charging, landing areas, passenger processing, and operational validation.¹³

Airport projects become less competitive when they rely only on broad market-growth forecasts. Forecast providers describe large UAM opportunity spaces, but they also segment the market across passenger transport, cargo logistics, medical and disaster response, private operations, charging stations, maintenance, repair and overhaul, air traffic control systems, and vertiports. That breadth helps justify airport planning, but it also means a proposed airport investment should specify which services it enables and which constraint it resolves. Passenger transport may be the leading application in some forecasts, while cargo, medical services, and emergency response are also identified as expanding UAM use cases. A stronger airport application therefore shows multi-use value rather than assuming air-taxi demand alone will carry the project.¹²¹³

The Orlando Geico Garage example illustrates why airport-adjacent funding requests need disciplined financial and structural assumptions. The conceptual downtown vertiport used a \$30 million retrofit assumption covering charging infrastructure, passenger and crew facilities, MRO facilities, vehicle gates, contingency, and structural reinforcement. Because the garage was not originally designed for roof-deck aviation use, the study stated that a detailed structural assessment would be needed to understand cost implications and retrofit risk. It also identified \$600,000 to \$1 million in estimated annual lost parking revenue, while suggesting a possible \$2.1 million annual ground lease that was not included in the IMPLAN inputs. The lesson is direct: competitiveness improves when applicants disclose displaced revenue, structural risk, cost-sharing assumptions, and lease or operator revenue separately from economic-impact modeling.⁴

A practical scoring standard follows from the evidence. The most competitive airport electrification and AAM projects should demonstrate a near-term operating concept, serve more than one end-user category, include charging and passenger or cargo handling, address airspace/traffic-management integration, and document local cost offsets and risks. Projects should be treated as weaker when they depend on unresolved autonomous operations, unspecified demand, or single-operator commercialization claims, because current sources still identify regulatory uncertainty, airspace complexity, infrastructure limitations, battery limits, and high initial deployment cost as active barriers.¹³

Legislative and regulatory recommendations

Legislation should treat AAM as a regulated mobility network, not a single air-taxi product. Market definitions already span air taxis, air metros, air ambulance, last-mile delivery, intercity and intracity service, piloted, autonomous, and hybrid operations, and end users including ridesharing companies, scheduled operators, e-commerce firms, hospitals, medical agencies, and private operators. This breadth argues for use-case-neutral rules that can support passenger, cargo, medical, and emergency-response operations without prematurely privileging one business model.¹¹

The first priority is a structured pilot-authority framework. Recent UAM market activity is being shaped by pilot programs, certification progress, infrastructure readiness, and airspace-integration validation. A useful statute should authorize limited deployments, require operator data sharing, specify safety and community-reporting obligations, and coordinate city, airport, state, and federal roles. This aligns with the observed need for policy mechanisms that let technology and regulation mature together, including UAS integration pilots, FAA rulemaking committees, and AAM/UAM challenge programs.¹³¹¹

Second, lawmakers should sequence autonomy rather than assume it. Current UAM deployments are still led by piloted operations because they fit existing regulatory frameworks, provide human oversight, support operational validation, and help build trust before autonomy. The UK's 2024 plan similarly framed passenger-carrying piloted eVTOL services for 2026, with autonomous trials expected later, by 2030. State and local rules should therefore enable piloted early operations while requiring a separate evidentiary showing for autonomous passenger service.¹³¹¹

Third, infrastructure permitting should be explicit. Vertiports and vertistops are described as essential for landing, charging, and passenger handling, and accounted for the largest UAM infrastructure share in 2026. Yet infrastructure gaps, high deployment costs, airspace integration, traffic-management complexity, and battery limitations remain deployment constraints. Local ordinances should require site plans covering charging, passenger processing, emergency access, structural capacity, noise/community impacts, and multimodal connections before public incentives or operating approvals are granted.¹³

Finally, public funding should be conditional. The Orlando vertiport case shows why: a conceptual garage retrofit assumed \$30 million in capital cost, included charging, passenger and crew facilities, MRO facilities, vehicle gates, and structural reinforcement contingency, and still required a detailed structural assessment outside the study scope. Legislation should require transparent cost-sharing, revenue-sharing, displacement-cost analysis, and public-benefit tests before public assets are converted to AAM use. This is especially important because legal, safety, privacy, investment, and regulatory barriers can slow or prevent widespread UAM adoption.⁴¹¹

Implementation Pathways

Technology, infrastructure, and regulatory assumptions

Implementation assumptions should treat AAM as an integrated mobility ecosystem, not an aircraft procurement exercise. The relevant system includes aircraft, MRO and ancillary services, operators, vertiport/airport infrastructure, airspace management, charging infrastructure, pilot training, ground handling, and certification. Market descriptions similarly place VTOL manufacturers, infrastructure developers, service providers, mobility operators, regulators, and city authorities in the same operating ecosystem.³¹³

Technology assumptions should remain architecture-specific. Multicopters are best matched to simple short urban trips, lift+cruise designs trade vertical lift for more efficient forward flight, and vectored-thrust designs offer greater speed and range with higher complexity. Battery capability remains a binding uncertainty: current lithium-ion cells are described at roughly 250–300 Wh/kg, while commercially viable operations are said to require 400–500+ Wh/kg. Early service assumptions should therefore avoid treating autonomy, long range, or high utilization as default conditions.³

Infrastructure assumptions should be explicit and site-level. Vertiports and vertistops are described as essential for landing, charging, and passenger handling, and MarketsandMarkets reports that this infrastructure segment accounted for the largest UAM infrastructure share in 2026. Retrofitting can carry material hidden costs: Orlando's Geico Garage concept used a \$30 million retrofit assumption covering charging, passenger and crew facilities, MRO facilities, vehicle gates, contingency, and likely structural reinforcement. That same case also identified foregone parking revenue of about \$600,000 to \$1 million annually, showing that public-site conversion has opportunity costs beyond construction.¹³⁴

Regulatory assumptions should be phased. Piloted operations currently dominate because they fit existing regulatory frameworks, provide human oversight for operational validation, and help build trust before autonomy. Certification and regulation are described as major determinants of market timing, with type certification proving more costly and time-consuming than expected across the industry. Airspace integration also remains a scaling constraint because urban operations require coordination among multiple aircraft and integration with existing airspace.¹³³

The most defensible planning baseline is therefore limited, piloted, short-range commercial service using certified aircraft, verified vertiport sites, dedicated charging capacity, conservative throughput assumptions, and public reporting on safety, noise, equity, and local economic impacts. Market forecasts and pilot announcements can inform scenarios, but they should not substitute for evidence that the aircraft, site, grid connection, airspace concept, operator economics, and regulatory approvals work together in a specific corridor.¹³³

Pilot programs and large UAM demonstrations

Pilot programs should be treated as evidence-gathering infrastructure, not proof that UAM is ready for broad deployment. NASA's AAM work has already framed demonstrations around system-level needs, including airspace design, vehicle development, community integration, fleet operations, and individual vehicle operations. This aligns with NASA's broader viability framework, where UAM depends on regulation, certification, cybersecurity, air traffic

management, infrastructure investment, affordability, willingness to pay, public acceptance, emissions, safety, communications, and traffic management rather than aircraft performance alone.³⁴²¹

The most useful early demonstrations are corridor-specific and constraint-specific. The Malibu-Century City route analysis shows that route choices can trade off duration, noise exposure, open-water safety requirements, Class D access, Class B constraints, helicopter-route congestion, and higher-speed energy use. The San Diego-Los Angeles reference mission shows the same pattern at regional scale: all studied departures required San Diego Class B entry and ATC routing, while alternative water, direct, inland, and coastal paths responded differently to noise, congestion, weather, restricted areas, and certification issues. Demonstrations should therefore test operating envelopes, not just passenger demand.⁶

Large demonstrations also need airspace and vertiport management objectives. Conventional controllers can manage limited air taxi volumes, but remotely piloted or autonomous high-volume operations require new UTM/U-space capabilities. HorizonUAM demonstrated a prototypical vertidrome manager integrated into a U-space environment, receiving real-time flight-plan, start-and-landing, emergency, and weather information in a scaled model-city test environment. That type of demonstration is valuable because scalable U-space functions can increase capacity, support on-demand flight planning, create no-fly zones, reduce noise and visual impacts, and improve societal acceptance.¹⁸

Regulatory trials should run ahead of permanent rule changes where possible. AAM regulatory actions can stretch more than 10 years, and airspace-change processes can take two to eight years depending on the change required. Rolling airspace trials are therefore a practical bridge: they can permit initial operations while longer airspace processes continue, give the public direct exposure to AAM services, improve investor confidence, and generate operational data for regulators.¹⁷

The near-term demonstration model should be piloted, limited, and data-rich. Several leading aircraft concepts described for 2025 still center on piloted service, including Joby's five-seat S4 and Archer's Midnight, while autonomous concepts such as Wisk's Generation 6 are positioned on a later certification path. Demonstrations should report route geometry, ATC interactions, vertiport throughput, charging turnaround, noise exposure, weather cancellations, emergency procedures, community feedback, and operating cost assumptions. Without those measurements, pilots risk becoming publicity events rather than decision-grade evidence for whether UAM can scale.³⁸

Operational constraint mitigation

Operational constraint mitigation should be treated as a staged systems exercise, not a single technical fix. The underlying ODM Aviation methodology identifies operational constraints by applying near-term ConOps to 12 Southern California reference missions, condensing 20 operational challenges into five constraints and three issues, then evaluating mitigation approaches by near-term implementability, network impact, ancillary effects, and scalability with denser operations. That framing matters: a mitigation that works for a few piloted flights may fail once operations become frequent, spatially dense, or politically visible.⁶

The first mitigation is route discipline. Early operations should use missions where demand, time value, infrastructure access, and constraint exposure are explicit. The Rancho Palos Verdes to Torrance Memorial Hospital mission illustrates the risk of assuming aviation value from urgency alone: the route is only 8.5 miles, can take 16 minutes off-peak and 28 minutes in peak traffic, and was not found to offer significant one-way time savings, though medical use may still create value because time can be critical and ODM could add consumer options. Medical and emergency pilots should therefore be designed as specialized service tests, not as evidence that short urban air taxi markets are broadly viable.⁶

The second mitigation is infrastructure matching. Los Angeles TOLA availability cannot be solved by one facility type; the cited study argues for a portfolio, including quieter and more maneuverable aircraft, improved community facilities, possible unimproved landing locations, and new TOLA infrastructure. Single-pad TOLAs are flexible, space-efficient, and comparatively cheap, but handle only one operation at a time; high-capacity TOLAs add parking, staging, services, and all-weather capabilities, but require more land, cost more, and concentrate noise and other impacts. This implies high-capacity sites should be reserved for airports, CBDs, event venues, and intermodal hubs, while low-volume corridors may need distributed single-pad access.⁶

The third mitigation is local planning before deployment. Regional agencies can address land use, zoning, route planning, charging, fleet storage, vertiport procedures, emergency procedures, fire suppression, passenger man-

agement, and building-code standards before operations scale. Public involvement should begin during long-range planning and continue through studies, pilots, demonstrations, and implementation because public acceptance is necessary for UAM to become a mobility option. NASA similarly identifies public acceptance as paramount and notes concerns over ground safety, noise, time of day, and operating altitude, making public interaction and testing in safe spaces central to risk reduction.^{[3440](#)}

The fourth mitigation is technology realism. NASA's eVTOL public-services white paper recommends matching Con-Ops to current technological maturity, while developing autonomous recharge/refuel logistics, hybrid systems, and other range-extending options in parallel. It also flags DEP infrastructure and certification delays as implementation challenges and calls for FAA coordination, operating-location infrastructure, and airspace-management preparation at the outset. Operational mitigation should therefore avoid assuming that battery maturity, autonomy, certification, vertiports, and airspace procedures will arrive simultaneously.^{[40](#)}

The practical pathway is incremental: select constrained but valuable corridors, define site-level TOLA and charging needs, run piloted demonstrations with explicit safety/noise/community reporting, and scale only after the same mitigation package works across infrastructure, airspace, public acceptance, and operating economics.

Community acceptance and proactive engagement

Community acceptance should be treated as an implementation requirement, not a communications add-on. Miami-Dade's UAM roadmap frames public confidence as crucial to UAM success and ties acceptance to strict safety and security standards, environmental protections, preservation of quality of life through noise and visual-impact controls, and avoidance of disproportionate impacts on any community or group. PennDOT similarly includes community acceptance and social equity among the factors used to guide AAM business-case development, alongside safety, air traffic management, environmental impacts, infrastructure readiness, climate, workforce, and economic potential.^{[3520](#)}

The engagement model therefore needs to begin before siting and continue through studies, pilots, demonstrations, and implementation. TBARTA recommends early public involvement during long-range transportation planning so agencies can prepare residents and learn which concerns matter locally; it also calls for ongoing community engagement to answer questions, gather feedback, and coordinate mitigations with the UAM sector. Miami-Dade identifies open houses, educational seminars, and focus groups as potential tools for an intentional, proactive public engagement strategy.^{[3435](#)}

Proactive engagement should also shape the regulatory and land-use process. Miami-Dade notes that planning must address airspace and operations, zoning, utility infrastructure, funding, social equity, and system connectivity, with siting analysis accounting for approach/departure airspace, adjacent building heights, noise-sensitive land uses, and potential harms to historically marginalized communities. PennDOT's regulatory strategy adds that agencies should clarify definitions, develop flexible frameworks, manage federal-state-local preemption tensions, collaborate across agencies and stakeholders, expand public education, and engage elected officials on national legislation.^{[3520](#)}

Public benefit should be explicit in any local approval or incentive package. Miami-Dade proposes incentives for OEMs and operators, such as early testing participation, input into vertiport siting, or reduced charging fees, in exchange for workforce development, discounted rides for low-income populations, and support for shared-mobility rather than exclusive private use. This turns acceptance from a reputational goal into a project condition: communities should see measurable commitments on safety, noise, access, equity, local jobs, and mitigation before AAM service is scaled.^{[35](#)}

Strategies to address discrepancies and inconsistencies

Discrepancies should be handled as planning inputs, not smoothed into a single market narrative. The Los Angeles reference missions show why: point-to-point ODM demand is not temporally or directionally consistent like commuting demand, and may appear sporadically across business, recreation, airport, medical, and private trips. The study's twelve reference missions were intentionally varied to expose different early adopter markets, constraints, and opportunities rather than prove one generalized use case.^{[6](#)}

The first strategy is to reconcile assumptions at the corridor level. San Diego to Los Angeles is a long-distance inter-city case: 122 miles, 110 minutes by car without congestion, and up to 240 minutes at peak, so the value proposition

is long nominal time reduction as much as congestion avoidance. Los Angeles City Center to Long Beach is different: travel times are stable until early afternoon, then show an evening congestion period with a 230% peak congestion penalty at 5:00 PM, implying demand concentrated in a narrow evening window. Beverly Hills Hotel to LAX is different again: the route has slow average speeds, large travel-time variance, airport-deadline sensitivity, and infrastructure gaps near both ends, including the closed LAX heliport. These cases should not share one utilization, willingness-to-pay, or service-frequency assumption.⁶

The second strategy is to separate “market attractiveness” from “operational readiness.” Beverly Hills Hotel to LAX may have wealthy customers, volatile ground congestion, and airport access value, but the same source identifies lack of nearby ground infrastructure and LAX heliport closure as limiting factors. Orlando’s Geico Garage example shows the same issue in economic form: a \$30 million retrofit assumption included charging, passenger and crew facilities, MRO, vehicle gates, and structural-reinforcement contingency, while foregone parking revenue was estimated at \$600,000 to \$1 million annually. A corridor should therefore pass separate screens for demand, site availability, retrofit cost, displaced revenue, charging needs, and operating permissions.⁶⁴

The third strategy is to use time-of-day and full-trip accounting. AAM can reduce travel time through direct aerial routing and congestion bypass, but it also affects modal choice, traffic decongestion, land use, infrastructure investment, equity, environmental impacts, and regulation. For short routes, access and terminal frictions can dominate; for long routes, en-route speed is more likely to matter. This distinction is consistent with the finding that passenger air travel is faster than road across distances but performs better on fuel cost and emissions only at longer distances.²⁶²⁸

Finally, every forecast should carry an uncertainty label: validated constraint, corridor-specific hypothesis, or unsupported extrapolation. Long-distance routes such as San Diego-Los Angeles and San Marino-Palm Springs were found to offer high-value time savings throughout the day, while Beverly Hills-LAX, LA-Long Beach, and Redondo Beach-Dodger Stadium were described as high-demand mainly during peak congestion, and Rancho Palos Verdes-Torrance Memorial Hospital did not show significant one-way time savings despite possible medical-service value. That hierarchy is the core discipline: preserve differences between missions until data justifies combining them.⁶

Conclusions and future research

The evidence supports a conditional conclusion: AAM is feasible only where aircraft design, mission profile, infrastructure, airspace, and economics are optimized together. Powertrain-aware optimization can reduce total direct operating cost by up to 6.6%, and its importance rises with mission range, but the dominant cost levers remain pilot cost, battery cost, battery manufacturing cost, and battery cycle life. The Los Angeles 30 km case also shows that short urban missions can be modeled in operational detail, yet even there results depend on boarding time, battery mass, bus voltage, inverter topology, and mission matching.¹⁵

The strongest planning implication is that route design is not a simple distance-minimization problem. In the San Diego-Los Angeles case, the direct route was shortest, but it introduced low-altitude overflight and congestion exposure near helicopter routes; the water route was 15% longer by distance and 11% longer by duration while reducing ground-noise exposure by operating 71% of its distance over open water. This confirms that future business cases should compare corridors by total system performance, not just flight time.⁶

Model uncertainty remains material. NDARC and SUAVE produced different estimates for GTOW, structural weight, cruise L/D, propulsive efficiency, hover figure of merit, and power loading for the same baseline aircraft, showing that conceptual design outputs should be treated as scenario-dependent rather than definitive. Infrastructure economics are similarly site-specific: the Orlando Geico Garage concept used a \$30 million retrofit assumption and identified \$600,000 to \$1 million in possible annual lost parking revenue, while noting that structural assessment was still outside scope.⁷⁴

Future research should focus on corridor-validated evidence: measured aircraft energy use and degradation by mission profile, powertrain thermal limits at longer range, vertiport retrofit costs after structural assessment, airspace throughput under realistic low-altitude separation, community response to repeated overflight, and integrated cost models that include access, boarding, charging, dispatch reliability, and lost alternative land uses. Until those data exist, AAM should be advanced through narrow, instrumented pilots rather than broad market assumptions.



Exhibit. Figure 19 Ferrovial and AECOM Vertiport Design. Source: S0051.

References

1. [Scientific assessment for urban air mobility \(UAM\) | CEAS Aeronautical Journal | Springer Nature Link](https://link.springer.com/academic). link.springer.com. academic. Source S0086.
2. [BAH_UAM_Executive Briefing](https://ntrs.nasa.gov/pdfs). ntrs.nasa.gov. pdf. Source S0042.
3. [The Global eVTOL and Advanced Air Mobility Market 2026-2036](https://giiresearch.com/standards_or_patent). giiresearch.com. standards_or_patent. Source S0080.
4. [Introduction As part of the City of Orlando Advanced Air Mobility \(AAM\) Transportation Plan, the VHB team prepared an Economic Impact Analysis of a conceptual downtown urban vertiport located at Geico Garage. The VHB team performed stakeholder interviews with Lilium, an electric vertical takeoff and landing \(eVTOL\) manufacturer, and researched publicly available information \(typically investment reports\) from other AAM manufacturers and operators to determine reasonable assumptions for capital costs, operating costs, and revenue projections. These assumptions were then presented during the NASA AAM Community Integration Workshop on December 15, 2021. The VHB team then collaborated with the Orlando Economic Partnership \(OEP\) to prepare an analysis of potential economic impact using the IMPLAN model. Please see Attachment A for the full technical memorandum summarizing the analysis. IMPLAN Input Assumptions: This study anticipated that the Geico Garage is part of a network of 14 total Vertiports throughout the State of Florida, including one vertiport at Lake Nona. Estimated revenue from passengers, capital cost, number of employees, and employment compensation were input into IMPLAN for analysis. The VHB team used \\$30M as a capital cost assumption to retrofit Geico Garage. This includes the electric charging infrastructure, passenger and crew facilities, maintenance repair and overhaul \(MRO\) facilities, and vehicle gates, and contingency for structural reinforcement. Because Geico Garage was not originally designed to support structures on the roof deck, structural reinforcements would likely be necessary. A detailed structural assessment is outside the scope of this assignment, and would be needed to fully understand the cost implications and risks associated with retrofitting the roof deck to vertiport use. No](#)

- [cost-sharing or revenue sharing agreements between the City and a development partner/AAM operator are assumed. Key City Considerations: VHB communicated with the City of Orlando Parking Division to determine the utilization of parking and potential revenue losses associated with converting the top level of the Geico Garage from its present use to vertiport use. Mr. Scott Zollars, Parking Division Manager, estimated approximately \\$600,000 to \\$1M of lost annual parking revenue \(assuming 100 events per year, 300 repurposed roof deck parking spaces, and a range of \\$20-\\$30 per space\). A parking lease agreement \(500 spaces per game\) with the Orlando Magic would also need to be re-negotiated. From: Curt Ostrodka, AICP, LEED AP Ref: 63588.02 April 19, 2022 Page 2 However, the annual lost parking revenue may be offset through a Ground Lease Agreement with an AAM operator. The analysis suggests that the City could lease the roof deck to an AAM operator at \\$2.1M/year, which would offset the parking revenue loss. This potential ground lease was not included as an input for the IMPLAN analysis. Economic Impact Analysis Results The economic impact assessment was completed using the IMPLAN input/output model to study impacts on all sectors and at all levels of an economy. IMPLAN uses input output analysis to estimate the spending patterns for every industry in a region. Table 1: IMPLAN Results. orlando.gov. pdf. Source S0003.](#)
5. [Beyond](#). secureenergy.org. pdf. Source S0069.
 6. [Parker Vascik dual masters thesis](#). seari.mit.edu. pdf. Source S0011.
 7. [Current Capabilities and Challenges of NDARC and SUAVE for eVTOL Aircraft Design and Analysis](#). suave.stanford.edu. pdf. Source S0007.
 8. <https://wsdot.wa.gov/sites/default/files/2025-06/Advanced-Air-Mobility-Aircraft-Plan-May2025.pdf>. ws-dot.wa.gov. pdf. Source S0038.
 9. [Urban Air Mobility Airspace Integration Concepts and Considerations](#). ntrs.nasa.gov. pdf. Source S0055.
 10. https://www.assureuas.com/wp-content/uploads/2021/06/A36-FinalReport_RevB.pdf. assureuas.com. pdf. Source S0061.
 11. [Urban Air Mobility \[UAM\] Market Size & Forecast Report, 2034](#). fortunebusinessinsights.com. company_or_market. Source S0111.
 12. [Urban Air Mobility Market Size, Share & Trends Report, 2040](#). rootsanalysis.com. academic. Source S0079.
 13. [Urban Air Mobility Market Report 2030-2035 \[427 Pages & 468 Tables\]](#). marketsandmarkets.com. standards_or_patent. Source S0082.
 14. [Preprint Manuscript | Forecasting Market Opportunities for Urban and Regional Air Mobility](#). ugpti.org. pdf. Source S0066.
 15. <http://chrismi.sdsu.edu/publications/227.pdf>. chrismi.sdsu.edu. pdf. Source S0008.
 16. <https://journal.hep.com.cn/fem/EN/PDF/10.1007/s42524-024-0298-0>. journal.hep.com.cn. pdf. Source S0117.
 17. [Advanced air mobility evidence review](#). assets.publishing.service.gov.uk. pdf. Source S0031.
 18. [Can Urban Air Mobility become reality? Opportunities and challenges of UAM as innovative mode of transport and DLR contribution to ongoing research | CEAS Aeronautical Journal | Springer Nature Link](#). link.springer.com. standards_or_patent. Source S0096.
 19. [eVTOL & Advanced Air Mobility Market Report 2026-2036 | Future Markets Inc.](#) futuremarketsinc.com. academic. Source S0078.
 20. https://www.pa.gov/content/dam/copapwp-pagov/en/penndot/documents/programs-and-doing-business/aviation/aviation-technology/penndot_aam_ae_feasibility_study_final_report.pdf. pa.gov. pdf. Source S0002.
 21. [URBAN AIR MOBILITY \(UAM\) MARKET STUDY](#). ntrs.nasa.gov. pdf. Source S0001.
 22. https://www.caee.utexas.edu/prof/kockelman/public_html/TPT_LCA_of_eVTOL.pdf. caee.utexas.edu. pdf. Source S0004.
 23. [\(PDF\) Operational Considerations regarding On-Demand Air Mobility: A Literature Review and Research Challenges](#). academia.edu. academic. Source S0024.

24. [Future Pathways for eVTOLs: A Design Optimization Perspective](#). arxiv.org. academic. Source S0060.
25. [From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Urban Air Mobility](#). arxiv.org. academic. Source S0059.
26. <https://apps.ict.illinois.edu/projects/getfile.asp?id=10857>. apps.ict.illinois.edu. pdf. Source S0012.
27. [eVTOL Passenger Acceptance](#). ntrs.nasa.gov. pdf. Source S0053.
28. [From Roads to Skies: The Next Revolution in](#). uam.berkeley.edu. pdf. Source S0005.
29. [Final Report Urban Air Mobility \(UAM\) Market Study](#). ntrs.nasa.gov. pdf. Source S0043.
30. [Urban Air Mobility Operational Concept \(OpsCon\) Passenger-Carrying Operations](#). ntrs.nasa.gov. pdf. Source S0039.
31. <https://oklahoma.gov/content/dam/ok/en/odot/about-us/boards-and-task-forces/Adv.%20Air%20Mob.%20Report%20OK%20Final%20EH%206.05.2024.pdf>. oklahoma.gov. pdf. Source S0034.
32. [Air Vehicle Design and Technology Considerations for an](#). adl2.stanford.edu. pdf. Source S0015.
33. [Advanced Air Mobility: What Electric Air Taxis Need to Take Off | Bain & Company](#). bain.com. standards_or_patent. Source S0091.
34. [Innovative Transit Technologies Study](#). canadianaam.com. pdf. Source S0050.
35. <https://miamidadetpo.org/library/studies/mdtpo-urban-air-mobility-policy-framework-and-strategic-roadmap-final-report-2023-11.pdf>. miamidadetpo.org. pdf. Source S0062.
36. <https://vtechworks.lib.vt.edu/server/api/core/bitstreams/e1f86645-229e-4046-a515-30fc7c364f3d/content>. vtechworks.lib.vt.edu. pdf. Source S0013.
37. [Integrating urban air mobility into the power grid through smart charging solutions](#). humnetlab.berkeley.edu. pdf. Source S0006.
38. [Urban Air Mobility 2025: Air-Taxi Pilot Programs, Launch Cities, and Regulatory Hurdles](#). ts2.tech. government. Source S0119.
39. [Urban Air Mobility Market Outlook 2026-2034: Market Share, and Growth Analysis by Vehicle Type, Operation, Application, Component](#). researchandmarkets.com. company_or_market. Source S0110.
40. https://ntrs.nasa.gov/api/citations/20205000636/downloads/2021-08-20-eVTOL-White-Paper-Final_V48.pdf. ntrs.nasa.gov. pdf. Source S0035.